Supporting Materials for "Automatic Differentiation to Improve Parameter Estimation for Partially Observed Markov Processes"

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Supplementary Content

S1 Proof of Theorem 1 (DMOP-α Functional Forms)

Theorem 1 follows immediately as a consequence of the following results, Lemmas S1 and S2. Here, an A superscript is used to denote ancestral trajectories, so that $x^{A,P,\theta}_{1:n,j}$ contains the history of the prediction particle $x^{P,\theta}_{n,j}$ at observation times $1:n$. Similarly, $g^{A,P,\theta}_{i,j}$ is the ancestral weight of $x^{P,\theta}_{n,j}$ at observation time i.

Lemma S1. *Write $\nabla_\theta \hat{\ell}^\alpha(\theta)$ for the DMOP-α gradient estimate (or, equivalently, the gradient of MOP-α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 1$,*

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J}\sum_{j=1}^J \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y^*_{1:N}|x^{A,F,\theta}_{1:n,j};\theta\big), \tag{S1}$$

yielding the bootstrap filter estimator of Poyiadjis et al. (2011) and Ścibior and Wood (2021).

Proof. Consider the case of MOP-α when $\alpha = 1$ and $\theta = \phi$. We then have a nice telescoping product property for the after-resampling likelihood estimate:

$$\begin{aligned}\hat{\mathcal{L}}^1(\theta) \quad &:= \quad \prod_{n=1}^N L_n^{A,\theta,\alpha} = \prod_{n=1}^N L_n^\phi \cdot \frac{\sum_{j=1}^J w^{F,\theta}_{n,j}}{\sum_{j=1}^J w^{P,\theta}_{n,j}} = \prod_{n=1}^N L_n^\phi \cdot \frac{\sum_{j=1}^J w^{F,\theta}_{n,j}}{\sum_{j=1}^J w^{F,\theta}_{n-1,j}} \\ &= \quad \left(\frac{1}{J}\sum_{j=1}^J w^{F,\theta}_{N,j}\right)\prod_{n=1}^N L_n^\phi,\end{aligned} \tag{S2}$$

where the third equality follows from the choice of $\alpha = 1$, and the fourth equality is the resulting telescoping property. The log-derivative identity lets us decompose the score estimate as

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{\nabla_\theta \hat{\mathcal{L}}^1(\theta)}{\hat{\mathcal{L}}^1(\theta)} = \frac{\nabla_\theta\left(\frac{1}{J}\sum_{j=1}^J w^{F,\theta}_{N,j}\right)\prod_{n=1}^N L_n^\phi}{\prod_{n=1}^N L_n^\phi} = \frac{1}{J}\sum_{j=1}^J \nabla_\theta w^{F,\theta}_{N,j}. \tag{S3}$$

From (S3), we see that the derivative of the log-likelihood estimate is

$$\nabla_\theta \hat{\ell}^1(\theta) := \frac{1}{J}\sum_{j=1}^J \nabla_\theta w^{F,\theta}_{N,j}. \tag{S4}$$

We proceed to decompose (S4). First, observe that as $\alpha = 1$,

$$w^{P,\theta}_{n,j} = w^{F,\theta}_{n-1,j}\frac{g^\theta_{n,j}}{g^\phi_{n,j}} = \prod_{i=1}^n \frac{g^{A,P,\theta}_{i,j}}{g^{A,P,\phi}_{i,j}}, \tag{S5}$$

Note that the right hand side of (S5) is the cumulative product of measurement density ratios over the ancestral trajectory of the j-th prediction particle at timestep n. We then use the log-derivative identity again, yielding the following expression for the gradient of the log-weights as the sum of the log measurement densities over the ancestral trajectory,

$$\frac{\nabla_\theta w^{P,\theta}_{n,j}}{w^{P,\theta}_{n,j}} = \nabla_\theta \log w^{P,\theta}_{n,j} \quad = \quad \nabla_\theta \log\left(\prod_{i=1}^n \frac{g^{A,P,\theta}_{i,j}}{g^{A,P,\phi}_{i,j}}\right) \tag{S6}$$

$$= \nabla_\theta \sum_{i=1}^{n} \left(\log g_{i,j}^{A,P,\theta} - \log g_{i,j}^{A,P,\phi}\right) \tag{S7}$$

$$= \sum_{i=1}^{n} \nabla_\theta \log g_{i,j}^{A,P,\theta}. \tag{S8}$$

This is equal to the gradient of the logarithm of the conditional density of the observed measurements given the ancestral trajectory of the j-th prediction particle up to timestep n,

$$\nabla_\theta \sum_{n=1}^{N} \log g_{n,j}^{A,P,\theta} = \nabla_\theta \log \left(\prod_{n=1}^{N} g_{n,j}^{A,P,\theta} \right) \tag{S9}$$

$$= \nabla_\theta \log \left(\prod_{n=1}^{N} f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{A,P,\theta};\theta\big) \right) \tag{S10}$$

$$= \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^*|x_{1:n,j}^{A,P,\theta};\theta\big). \tag{S11}$$

Multiplying both sides of the expression by $w_{N,j}^{P,\theta}$ yields an expression for the gradient of the weights at timestep N,

$$\nabla_\theta w_{N,j}^{P,\theta} = w_{N,j}^{P,\theta} \sum_{n=1}^{N} \nabla_\theta \log g_{n,j}^{A,P,\theta} = w_{N,j}^{P,\theta} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^*|x_{1:n,j}^{A,P,\theta};\theta\big). \tag{S12}$$

Substituting the above identity into the log-likelihood decomposition obtained earlier in Equation S3 yields

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta w_{N,j}^{F,\theta} = \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta w_{N,k_j}^{P,\theta}$$

$$= \frac{1}{J}\sum_{j=1}^{J} w_{N,k_j}^{P,\theta} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^*|x_{1:n,k_j}^{A,P,\theta};\theta\big). \tag{S13}$$

Finally, observing that $\theta = \phi$ implies $w_{N,j}^{F,\theta} = 1$, we obtain

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^*|x_{1:n,j}^{A,F,\theta};\theta\big). \tag{S14}$$

This has the same undiscounted path-space score target as the estimators of Poyiadjis et al. (2011); Ścibior and Wood (2021) for the bootstrap filter. The reparameterization score contribution need not equal their complete-data likelihood-ratio score contribution particle by particle. □

Note that the variance of the DMOP-α log-likelihood derivative estimate scales poorly with N when $\theta \neq \phi$. This can be seen by observing that

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta w_{N,j}^{F,\theta} = \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta w_{N,k_j}^{P,\theta}$$

$$= \quad \frac{1}{J}\sum_{j=1}^{J} w^{P,\theta}_{N,k_j} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y^*_{1:N}|x^{A,P,\theta}_{1:n,k_j};\theta\big). \tag{S15}$$

When $\theta \neq \phi$, we see that $w^{P,\theta}_{N,k_j} = O(c^N)$. When $\theta = \phi$, a generic normalized path-functional inequality gives the coarse upper bound $O(N^4/J)$ after rescaling to the full unnormalized score. Section S7 gives an exact constant-weight calculation with order N^2/J at $\alpha = 1$ and explains why this example does not by itself provide a general finite-J upper bound.

Lemma S2. *Write $\nabla_\theta \hat{\ell}^\alpha(\theta)$ for the DMOP-α gradient estimate (or, equivalently, the gradient of MOP-α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 0$,*

$$\nabla_\theta \hat{\ell}^0(\theta) = \frac{1}{J}\sum_{n=1}^{N}\sum_{j=1}^{J} \nabla_\theta \log\Big(f_{Y_n|X_n}\big(y^*_n|x^{F,\theta}_{n,j};\theta\big)\Big), \tag{S16}$$

yielding the estimate of Naesseth et al. (2018) when applied to the bootstrap filter.

Proof. First, write

$$s_{n,j} = \frac{f_{Y_n|X_n}\big(y^*_n|x^{P,\theta}_{n,j};\theta\big)}{f_{Y_n|X_n}\big(y^*_n|x^{P,\phi}_{n,j};\theta\big)} \tag{S17}$$

as shorthand for the measurement density ratios. Observe that, when $\alpha = 0$,, the likelihood estimate becomes

$$\begin{aligned}
\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^N L_n^{A,\theta,\alpha} \quad &= \quad \prod_{n=1}^N L_n^\phi \cdot \frac{\sum_{j=1}^J w^{F,\theta}_{n,j}}{\sum_{j=1}^J w^{P,\theta}_{n,j}} && \text{(S18)}\\
&= \quad \prod_{n=1}^N L_n^\phi \cdot \frac{1}{J}\sum_{j=1}^J s_{n,j} && \text{(S19)}\\
&= \quad \prod_{n=1}^N L_n^\phi \cdot \frac{1}{J}\sum_{j=1}^J \frac{f_{Y_n|X_n}\big(y^*_n|x^{P,\theta}_{n,j};\theta\big)}{f_{Y_n|X_n}\big(y^*_n|x^{P,\phi}_{n,j};\theta\big)}. && \text{(S20)}
\end{aligned}$$

We lose the nice telescoping property observed in the MOP-1 case, but this expression still yields something useful. This is because its gradient when $\theta = \phi$ is therefore

$$\begin{aligned}
\nabla_\theta \hat{\ell}^0(\theta) := \sum_{n=1}^N \nabla_\theta \log\left(L_n^\phi \frac{1}{J}\sum_{j=1}^J s_{n,j}\right) && \text{(S21)}\\
= \sum_{n=1}^N \frac{\nabla_\theta\left(L_n^\phi \frac{1}{J}\sum_{j=1}^J s_{n,j}\right)}{\left(L_n^\phi \frac{1}{J}\sum_{j=1}^J s_{n,j}\right)} && \text{(S22)}\\
= \sum_{n=1}^N \frac{\sum_{j=1}^J \nabla_\theta s_{n,j}}{\sum_{j=1}^J s_{n,j}} && \text{(S23)}
\end{aligned}$$

$$= \sum_{n=1}^{N} \frac{1}{J} \sum_{j=1}^{J} \frac{\nabla_\theta f_{Y_n|X_n}\left(y_n^*|x_{n,j}^{F,\theta};\theta\right)}{f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\phi};\phi)} \tag{S24}$$

$$= \frac{1}{J} \sum_{n=1}^{N} \sum_{j=1}^{J} \nabla_\theta \log\left(f_{Y_n|X_n}\left(y_n^*|x_{n,j}^{F,\theta};\theta\right)\right), \tag{S25}$$

where we use the log-derivative trick in the second equality, observe that $\sum_{j=1}^{J} s_{n,j} = J$ when $\theta = \phi$ in the fourth equality, and use the log-derivative trick again while noting that $\theta = \phi$ in the fifth equality. This yields the desired result. □

S2 Proof of Theorem 2 (Optimization Convergence Analysis)

All derivatives in this section are evaluated at the common filtering and evaluation parameter $\theta = \phi$. Let Z_n contain the complete ancestral path through time n, the parameter-independent simulator variables, and the first- and second-order gradients propagated by automatic differentiation. Set

$$G_n^\theta(Z_n) = f_{Y_n|X_n}(y_n^* \mid X_n^\theta;\theta), \qquad W_\theta(Z_N) = \prod_{n=1}^{N} G_n^\theta(Z_n), \tag{S26}$$

$$\psi_\theta(Z_N) = \nabla_\theta^{\mathrm{tot}} \log W_\theta(Z_N), \qquad \Xi_\theta(Z_N) = \nabla_\theta^{2,\mathrm{tot}} \log W_\theta(Z_N). \tag{S27}$$

The superscript "tot" means that the derivative passes through the reparameterized simulated path while the base random numbers and the discrete resampling indices are held fixed.

Write $\rho_N^J = J^{-1}\sum_{j=1}^{J} \delta_{Z_{N,j}}$ for the empirical measure of the after-resampling genealogical particles, and define the normalized Feynman–Kac path measure by

$$\rho_N(F) = \frac{\mathbb{E}\{W_\theta(Z_N)F(Z_N)\}}{\mathbb{E}\{W_\theta(Z_N)\}}. \tag{S28}$$

Assumptions (A3) and (A5) give $|\partial_{\theta_i} W_\theta| \le N\bar{G}^N G'(\theta)$, so dominated convergence and the log-derivative identity yield

$$\nabla_\theta \ell(\theta) = \rho_N(\psi_\theta), \qquad \nabla_\theta \hat{\ell}_J^{A,1}(\theta) = \rho_N^J(\psi_\theta). \tag{S29}$$

The second equality is Lemma S1; importantly, it is one terminal empirical average of the complete path score, not a sum of empirical filtering expectations at the individual observation times.

For completeness, multinomial resampling gives the required genetic Feynman–Kac particle system explicitly. For semigroup indexing, let Z_0 be the augmented prediction path at observation 1; for $q = 1, \ldots, N-1$, let Z_q be the prediction path at observation $q+1$; and let Z_N be the terminal path after the final resampling. For $0 \le q \le N$, write $\rho_q^J = J^{-1}\sum_{j=1}^{J} \delta_{\mathsf{Z}_{q,j}}$, where the $\mathsf{Z}_{0,j}$ are independent draws from the unweighted prediction law of Z_0 and $\mathsf{Z}_{N,j} = Z_{N,j}$. Write $V_q^\theta(\mathsf{Z}_q) = G_{q+1}^\theta(\mathsf{Z}_q)$ for $q = 0, \ldots, N-1$. Let the locally indexed augmented kernel $\mathcal{M}_{q+1}^\theta$ append the next independent simulator innovation and its gradients for $0 \le q \le N-2$, and put $\mathcal{M}_N^\theta = \mathrm{Id}$ for the final selection. The one-step population map is

$$\Phi_{q+1}^\theta(\mu)(dz') = \frac{\int \mu(dz) V_q^\theta(z) \mathcal{M}_{q+1}^\theta(z, dz')}{\mu(V_q^\theta)}, \qquad 0 \le q < N. \tag{S30}$$

Conditionally on the current cloud, the J children are independent with common law $\Phi^\theta_{q+1}(\rho^J_q)$. The final observation is handled by the $q = N-1$ transition, namely multinomial selection with potential $V^\theta_{N-1} = G^\theta_N$ followed by the identity mutation. Thus ρ^J_N is exactly the terminal empirical measure of this augmented historical system. Systematic resampling does not have this conditional-independence property and is not covered by the calculation.

We next write out the particle concentration constant used below. For $0 \le q \le N$, let

$$Q^\theta_{q,N}F(z_q) = \mathbb{E}\left[F(\mathsf{Z}_N) \prod_{r=q}^{N-1} V^\theta_r(\mathsf{Z}_r) \,\middle|\, \mathsf{Z}_q = z_q \right], \tag{S31}$$

$$P^\theta_{q,N}F = \frac{Q^\theta_{q,N}F}{Q^\theta_{q,N}1}, \qquad g_{q,N} = \sup_{z,z'} \frac{Q^\theta_{q,N}1(z)}{Q^\theta_{q,N}1(z')}, \qquad b_{q,N} = \sup_{\operatorname{osc}(F)\le 1} \operatorname{osc}(P^\theta_{q,N}F), \tag{S32}$$

where $Q^\theta_{N,N}F = F$ and $\operatorname{osc}(F) = \sup_{z,z'} |F(z) - F(z')|$. Thus $0 \le b_{q,N} \le 1$. With

$$R_G = \frac{\bar{G}}{\underline{\mathrm{G}}}, \qquad S_N = \sum_{q=0}^{N} g^3_{q,N} b_{q,N}, \qquad K_N = 32 S_N, \qquad K^{\mathrm{pf}}_N = 32 \sum_{r=0}^{N} R_G^{3r}, \tag{S33}$$

the potential bounds give, term by term,

$$\underline{\mathrm{G}}^{N-q} \le Q^\theta_{q,N}1(z_q) \le \bar{G}^{N-q}, \qquad g_{q,N} \le R_G^{N-q}, \qquad K_N \le K^{\mathrm{pf}}_N. \tag{S34}$$

In particular, the constant used in the theorem is the closed-form quantity

$$K^{\mathrm{pf}}_N = \begin{cases} 32(N+1), & R_G = 1, \\ 32\dfrac{R_G^{3(N+1)} - 1}{R_G^3 - 1}, & R_G > 1. \end{cases} \tag{S35}$$

Here is the constant calculation behind the tail bound used below. Let r^{DM}_N and β^{DM}_N denote the second- and first-order coefficients in Theorem 1.2 of Del Moral and Rio (2011). The Feynman–Kac bounds in their Appendix A.3 give

$$r^{\mathrm{DM}}_N \le 4S_N, \qquad (\beta^{\mathrm{DM}}_N)^2 \le 4\sum_{q=0}^{N} g^2_{q,N} b^2_{q,N} \le 4S_N, \tag{S36}$$

where the last inequality uses $g_{q,N} \ge 1$ and $0 \le b_{q,N} \le 1$. If $S_N = 0$, the particle error is identically zero. Hence suppose $S_N > 0$. For $u \in (0, 1/2]$, set $x = Ju^2/(32S_N)$. If $x \le \log 6$, then $6e^{-x} \ge 1$ and the desired probability bound is trivial. If $x > \log 6$, use $h^{-1}(x) \le 2x + 2\sqrt{x}$ for $h(t) = \{t - \log(1+t)\}/2$ and $1 + 2x + 2\sqrt{x} \le 8x$ to obtain

$$\frac{r^{\mathrm{DM}}_N}{J}\{1 + h^{-1}(x)\} \le \frac{u^2}{8x}\{1 + 2x + 2\sqrt{x}\} \le u^2 \le \frac{u}{2},$$
$$\beta^{\mathrm{DM}}_N \sqrt{\frac{2x}{J}} \le 2\sqrt{S_N}\sqrt{\frac{2x}{J}} = \frac{u}{2}.$$

Theorem 1.2 applied to both F and $-F$, after centering and scaling F to oscillation one, therefore gives the explicit Feynman–Kac corollary

$$\Pr\left(|(\rho_N^J - \rho_N)(F)| > \operatorname{osc}(F)u\right) \le 6\exp\left(-\frac{Ju^2}{K_N}\right) \le 6\exp\left(-\frac{Ju^2}{K_N^{\mathrm{pf}}}\right). \tag{S37}$$

for every F with $\operatorname{osc}(F) > 0$ and every $u \in (0, 1/2]$. If F is constant, $(\rho_N^J - \rho_N)(F) = 0$ identically. Equations S33– S37 replace all unnamed model-dependent concentration constants.

Lemma S3 (Explicit concentration of the DMOP-1 gradient)**.** *Assume multinomial resampling and let*

$$A_N(\theta) = \max_{1\le i\le p} \operatorname{osc}(\psi_{\theta,i}), \qquad u_g(J,\delta) = \sqrt{\frac{K_N^{\mathrm{pf}}}{J}\log\frac{6p}{\delta}}. \tag{S38}$$

If $J \ge 4K_N^{\mathrm{pf}}\log(6p/\delta)$, then, with probability at least $1-\delta$,

$$\|\nabla_\theta \hat\ell_J^{A,1}(\theta) - \nabla_\theta \ell(\theta)\|_2 \le A_N(\theta)\sqrt{p}\, u_g(J,\delta). \tag{S39}$$

Moreover, Assumption (A3) gives $A_N(\theta) \le 2NG'(\theta)$. Consequently, the explicit condition

$$J \ge K_N^{\mathrm{pf}} \log\frac{6p}{\delta} \max\left\{4, \frac{4pN^2G'(\theta)^2}{\epsilon^2}\right\} \tag{S40}$$

implies that the left side of equation S39 is at most ϵ with probability at least $1-\delta$.

Proof. The lower bound on J makes $u_g(J,\delta) \le 1/2$. Apply equation S37 to each normalized coordinate $\psi_{\theta,i}/A_N(\theta)$; a constant coordinate has zero error and needs no normalization. A union bound over the p coordinates gives

$$\max_{1\le i\le p} |(\rho_N^J - \rho_N)(\psi_{\theta,i})| \le A_N(\theta)u_g(J,\delta)$$

with probability at least $1-\delta$. Now use $\|v\|_2 \le \sqrt{p}\|v\|_\infty$ and equation S29. Finally, $|\psi_{\theta,i}| \le NG'(\theta)$, so $A_N(\theta) \le 2NG'(\theta)$; substituting this inequality and solving $2NG'(\theta)\sqrt{p}\, u_g(J,\delta) \le \epsilon$ gives equation S40. $\square$

The next result treats the actual particle Hessian. It retains the outer-product and empirical-mean terms created by differentiating the logarithm of the terminal empirical average.

Lemma S4 (Explicit concentration of the DMOP-1 particle Hessian)**.** *Suppose in addition that Assumption (A3) gives the entrywise bound $|\Xi_{\theta,ab}(Z_N)| \le NG''(\theta)$ for $1 \le a, b \le p$. Define*

$$d_p = \frac{p(p+3)}{2}, \qquad H_N(\theta) = 2p\{NG''(\theta) + 3N^2G'(\theta)^2\}, \qquad u_H(J,\delta) = \sqrt{\frac{K_N^{\mathrm{pf}}}{J}\log\frac{6d_p}{\delta}}. \tag{S41}$$

If $J \ge 4K_N^{\mathrm{pf}}\log(6d_p/\delta)$, then, with probability at least $1-\delta$,

$$\|\nabla_\theta^2 \hat\ell_J^{A,1}(\theta) - \nabla_\theta^2 \ell(\theta)\|_{\mathrm{op}} \le H_N(\theta)u_H(J,\delta). \tag{S42}$$

Fix $0 < \kappa < \gamma$. *If*

$$\gamma I_p \preceq -\nabla^2_\theta \ell(\theta) \preceq \Gamma I_p \quad \textit{and} \quad J \geq K_N^{\mathrm{pf}} \log \frac{6d_p}{\delta} \max\left\{4, \frac{H_N(\theta)^2}{\kappa^2}\right\}, \tag{S43}$$

then, with probability at least $1-\delta$,

$$(\gamma - \kappa) I_p \preceq -\nabla^2_\theta \hat{\ell}_J^{A,1}(\theta) \preceq (\Gamma + \kappa) I_p. \tag{S44}$$

In particular, the raw particle curvature is then positive definite and invertible.

Proof. For fixed reference-filter random numbers and resampling indices, the telescoping identity in the proof of Lemma S1 gives

$$\hat{\ell}_J^{A,1}(\theta) = C_\phi + \log\left\{\frac{1}{J}\sum_{j=1}^{J} \exp(a_j(\theta))\right\}, \qquad a_j(\theta) = \log w_{N,j}^{F,\theta}, \qquad a_j(\phi) = 0,$$

where C_ϕ is constant in θ. Twice differentiating this log-sum-exp at $\theta = \phi$ gives the exact finite-particle identity

$$\nabla^2_\theta \hat{\ell}_J^{A,1} = \rho_N^J(\Xi_\theta + \psi_\theta \psi_\theta^T) - \rho_N^J(\psi_\theta)\rho_N^J(\psi_\theta)^T. \tag{S45}$$

Assumption (A3) also gives the entrywise dominating bound

$$|\partial^2_{\theta_a\theta_b} W_\theta| = W_\theta |\Xi_{\theta,ab} + \psi_{\theta,a}\psi_{\theta,b}| \leq \bar{G}^N\{NG''(\theta) + N^2 G'(\theta)^2\}.$$

Thus differentiation through second order may pass under the expectation, and differentiating $\log \mathbb{E} W_\theta$ gives the population identity

$$\nabla^2_\theta \ell = \rho_N(\Xi_\theta + \psi_\theta \psi_\theta^T) - \rho_N(\psi_\theta)\rho_N(\psi_\theta)^T. \tag{S46}$$

Let $A_1 = 2NG'(\theta)$ and $A_2 = 2\{NG''(\theta) + N^2 G'(\theta)^2\}$. The p scalar functions ψ_a have oscillation at most A_1, and the symmetric entries of $\Xi + \psi\psi^T$ have oscillation at most A_2. Apply equation S37 jointly to these functions, using $d_p = p + p(p+1)/2$ scalar events. On the resulting event,

$$\|(\rho_N^J - \rho_N)(\psi)\|_2 \leq A_1 \sqrt{p}\, u_H(J,\delta),$$
$$\|(\rho_N^J - \rho_N)(\Xi + \psi\psi^T)\|_{\mathrm{op}} \leq p A_2 u_H(J,\delta).$$

Both $\|\rho_N^J(\psi)\|_2$ and $\|\rho_N(\psi)\|_2$ are at most $\sqrt{p}\, NG'(\theta)$. Therefore

$$\begin{aligned}
&\|\rho_N^J(\psi)\rho_N^J(\psi)^T - \rho_N(\psi)\rho_N(\psi)^T\|_{\mathrm{op}} \\
&\qquad \leq 2\sqrt{p}\, NG'(\theta)\, \|(\rho_N^J - \rho_N)(\psi)\|_2 \leq 4pN^2 G'(\theta)^2 u_H(J,\delta).
\end{aligned}$$

Combining the last three displays with equation S45 and equation S46 proves equation S42. Condition equation S43 makes its right side at most κ; Weyl's eigenvalue inequality then gives equation S44. □

For the optimization theorem, no stochastic Hessian claim is needed if the curvature is regularized deterministically. If $\widetilde{B} = Q\,\mathrm{diag}(\lambda_1,\ldots,\lambda_p)Q^T$ is an orthogonal eigendecomposition of any symmetric raw estimate, define

$$B = Q\,\mathrm{diag}([\lambda_1]_c^C,\ldots,[\lambda_p]_c^C)Q^T, \qquad [x]_c^C = \min\{C,\max\{c,x\}\}. \tag{S47}$$

For every $v \in \mathbb{R}^p$, direct expansion in the eigenbasis gives

$$c\|v\|_2^2 \le v^T B v \le C\|v\|_2^2, \qquad C^{-1}\|v\|_2^2 \le v^T B^{-1} v \le c^{-1}\|v\|_2^2. \tag{S48}$$

Thus $cI_p \preceq B \preceq CI_p$ and $C^{-1}I_p \preceq B^{-1} \preceq c^{-1}I_p$. Lemma S4 shows that the unclipped particle Hessian has these properties with $c = \gamma - \kappa$ and $C = \Gamma + \kappa$ under equation S43; equation S47 guarantees the chosen c, C bounds for every realization.

Proof of Theorem 2. Let τ and $\mathcal{E}_M$ be the stopping time and simultaneous gradient event defined in Theorem 2. If $\mathcal{F}_m$ is the history before the mth particle-filter call, then θ_m and $\{m \le \min(\tau, M-1)\}$ are $\mathcal{F}_m$-measurable. Lemma S3, with failure probability δ/M, gives

$$\begin{aligned}\Pr(\mathcal{E}_M^c \mid \theta_0) &\le \sum_{m=0}^{M-1} \mathbb{E}\left[\mathbf{1}_{\{m\le\min(\tau,M-1)\}} \Pr\left(\|g_m - \nabla f(\theta_m)\|_2 > \epsilon \mid \mathcal{F}_m\right) \,\middle|\, \theta_0\right] \\ &\le \sum_{m=0}^{M-1} \frac{\delta}{M} = \delta.\end{aligned} \tag{S49}$$

This calculation is valid at the adaptive iterates because each call uses a fresh particle-randomness block.

Work on $\mathcal{E}_M$ and write $p_m = -B_m^{-1} g_m$. If $\|g_m\|_2 \le \sigma\epsilon$, then

$$\|\nabla f(\theta_m)\|_2 \le \|g_m\|_2 + \epsilon \le (\sigma+1)\epsilon. \tag{S50}$$

Strong convexity implies $f(\theta) - f(\theta^*) \le \|\nabla f(\theta)\|_2^2/(2\gamma)$; therefore

$$f(\theta_m) - f(\theta^*) \le \frac{(\sigma+1)^2\epsilon^2}{2\gamma}. \tag{S51}$$

Suppose instead that $\|g_m\|_2 > \sigma\epsilon$. Since $g_m = -B_m p_m$ and $cI_p \preceq B_m \preceq CI_p$,

$$c\|p_m\|_2 \le \|g_m\|_2 \le C\|p_m\|_2, \qquad \epsilon\|p_m\|_2 < \frac{C}{\sigma}\|p_m\|_2^2. \tag{S52}$$

The smoothness inequality and the gradient-error event give

$$\begin{aligned}f(\theta_m + \eta p_m) - f(\theta_m) &\le \eta\nabla f(\theta_m)^T p_m + \frac{\Gamma\eta^2}{2}\|p_m\|_2^2 \\ &\le -\eta p_m^T B_m p_m + \eta\epsilon\|p_m\|_2 + \frac{\Gamma\eta^2}{2}\|p_m\|_2^2.\end{aligned} \tag{S53}$$

The choices of σ and η imply

$$\frac{C}{\sigma}\|p_m\|_2^2 \le \frac{c(1-\beta)}{2}\|p_m\|_2^2, \qquad \frac{\Gamma\eta}{2}\|p_m\|_2^2 \le \frac{c(1-\beta)}{2}\|p_m\|_2^2.$$

Using $p_m^T B_m p_m \ge c\|p_m\|_2^2$ in equation S53 now gives

$$f(\theta_{m+1}) - f(\theta_m) \le -\eta\beta p_m^T B_m p_m. \tag{S54}$$

On a nonstopping iteration,

$$\|\nabla f(\theta_m)\|_2 \le \|g_m\|_2 + \epsilon < \frac{\sigma+1}{\sigma}\|g_m\|_2. \tag{S55}$$

Consequently, equation S48 and strong convexity give

$$\begin{aligned} p_m^T B_m p_m &= g_m^T B_m^{-1} g_m \\ &\ge \frac{1}{C}\left(\frac{\sigma}{\sigma+1}\right)^2 \|\nabla f(\theta_m)\|_2^2 \\ &\ge \frac{2\gamma}{C}\left(\frac{\sigma}{\sigma+1}\right)^2 \{f(\theta_m) - f(\theta^*)\}. \end{aligned} \tag{S56}$$

Combining equation S54 and equation S56 proves

$$f(\theta_{m+1}) - f(\theta^*) \le q\{f(\theta_m) - f(\theta^*)\}, \qquad q = 1 - \frac{2\eta\beta\gamma}{C}\left(\frac{\sigma}{\sigma+1}\right)^2. \tag{S57}$$

The step-size bound also verifies the claimed range of the contraction:

$$0 < 1 - q < 2\beta(1-\beta)\frac{c\gamma}{C\Gamma} \le \frac{1}{2}, \qquad \text{so} \qquad \frac{1}{2} < q < 1.$$

Iteration gives the q^m conclusion in the theorem, while equation S51 gives its stopping conclusion after substituting $f = -\ell$. □

The result is deliberately restricted to DMOP-1. For $\alpha < 1$, the population limit is a discounted gradient. A theorem for that case must add a finite-sample bound for the discount bias relative to the exact score; choosing α close to one does not by itself provide such a bound.

S3 Feynman-Kac Models and Monte Carlo Approximations

In this section, we introduce the Feynman-Kac convention of Del Moral (2004) that has since become commonplace (Karjalainen et al., 2023) for the analysis of the particle filter. The mathematical formalization and notation introduced here will be adopted in the remainder of the analysis, in order to prove Theorems 3, 4, and 5. Let $(\eta_n)_{n=1}^N, (\pi_n)_{n=1}^N, (\rho_n)_{n=1}^N$ be sequences of probability measures on the state space $\mathcal{X}$. This is the sequence of prediction distributions $f_{X_n|Y_{1:n-1}}$, filtering distributions $f_{X_n|Y_{1:n}}$, and posterior distributions $f_{X_{1:n}|Y_{1:n}}$ that we seek to approximate with the particle filter. For any measurable bounded functional h, we adopt the following functional-analytic notation, borrowed from Del Moral (2004); Chopin and Papaspiliopoulos (2020); Karjalainen et al. (2023). We choose our specific choice of notation and definitions to be in line with that of Karjalainen et al. (2023).

Markov kernels and the process model: A Markov kernel M with source $\mathcal{X}_1$ and target $\mathcal{X}_2$ is a map $M : \mathcal{X}_1 \times \mathcal{B}(\mathcal{X}_2) \to [0, 1]$ such that for every set $A \in \mathcal{B}(\mathcal{X}_2)$ and every point $x \in \mathcal{X}_1$, the map $x \mapsto M(x, A)$ is a measurable function of x, and the map $A \mapsto M(x, A)$ is a probability measure on $\mathcal{X}_2$. The quantity $M(x, A)$ can be thought of as the probability of transitioning to the set A given that we are at the point x. If this yields a density, this then corresponds to the process density $f_{X_2|X_1}$ conditional on x and integrated over A.

Markov kernels and measures: For any measure η, any Markov kernel M on $\mathcal{X}$, any point $x \in \mathcal{X}$ and any measurable subset $A \subseteq \mathcal{X}$, let

$$\eta(h) = \int h\, d\eta = \int h(x)\eta(dx), \tag{S58}$$

$$(\eta M)(A) = \int \eta(dx)M(x, A), \tag{S59}$$

$$(Mh)(x) = \int M(x, dy)h(y). \tag{S60}$$

Compositions of Markov kernels: The composition of a Markov kernel M_1 with another Markov kernel M_2 is another Markov kernel, given by

$$(M_1M_2)(x, A) = \int M_1(x, dy)M_2(y, A). \tag{S61}$$

Total variation distance: The total variation distance between two measures μ and ν on $\mathcal{X}$ is

$$\|\mu - \nu\|_{\mathrm{TV}} = \sup_{\|h\|_\infty \le 1/2} |\mu(\phi) - \nu(\phi)| = \sup_{\mathrm{osc}(h) \le 1} |\mu(h) - \nu(h)|. \tag{S62}$$

Dobrushin contraction: The Dobrushin contraction coefficient β_{TV} of a Markov kernel M is given by

$$\beta_{\mathrm{TV}}(M) = \sup_{x,y \in \mathcal{X}} \big\|M(x, \cdot) - M(y, \cdot)\big\|_{\mathrm{TV}} = \sup_{\mu,\nu \in \mathcal{P}, \mu \ne \nu} \frac{\|\mu M - \nu M\|_{\mathrm{TV}}}{\|\mu - \nu\|_{\mathrm{TV}}}. \tag{S63}$$

Potential functions and the measurement model: A potential function $G : \mathcal{X} \to [0, \infty)$ is a non-negative function of an element of the state space $x \in \mathcal{X}$. In our case, this corresponds to the measurement model, and in our previous notation is written as $g_{n,j} = f_{Y_n|X_n}\big(y_n^*|x_{n,j}^F; \theta\big) = G_n(x_{n,j}^F)$, where we suppress the dependence on θ for notational simplicity. Note that $G_n(\cdot) = f_{Y_n|X_n}(y_n^*|\cdot)$ is the conditional density of the observed measurement at time n, where we condition on the filtering particle $x_{n,j}^F$ as an element of the state space.

Feynman-Kac models: A Feynman-Kac model on $\mathcal{X}$ is a tuple $(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N)$ of an initial probability measure on the state space π_0, a sequence of transition kernels $(M_n)_{n=1}^N$, and a sequence of potential functions $(G_n)_{n=1}^N$. In the notation used in the main text, this corresponds to the starting distribution f_{X_0}, the sequence of transition densities $f_{X_n|X_{n-1}}$, and the measurement densities $f_{Y_n|X_n}$. This induces a set of mappings from the set of probability measures on $\mathcal{X}$ to itself, $\mathcal{P}(\mathcal{X}) \to \mathcal{P}(\mathcal{X})$, as follows:

- The update from the prediction to the filtering distributions is given by

$$\pi_n(dx) = \Psi_n(\eta_n)(dx) = \frac{G_n(x) \cdot \eta_n(dx)}{\eta_n(G_n)}. \tag{S64}$$

- The map from the prediction distribution at timestep n to timestep $n+1$ is given by

$$\Phi_{n+1}(\eta_n) = \Psi_n(\eta_n) M_{n+1}. \tag{S65}$$

- The composition of maps between prediction distributions yields the map from the prediction distribution at time k to the prediction distribution at time n where $k \leq n$,

$$\Phi_{k,n} = \Phi_n \circ ... \circ \Phi_{k+1}. \tag{S66}$$

The particle filter: The particle filter then yields a Monte Carlo approximation to the above Feynman-Kac model, via a sequence of mixture Dirac measures. When one resamples at every timestep, the prediction measure at timestep n is then given by

$$\eta_n^J = \frac{1}{J} \sum_{j=1}^{J} \delta_{x_{n,j}^P}, \tag{S67}$$

and the filtering measure at timestep n is given by

$$\pi_n^J = \frac{\sum_{j=1}^J g_{n,j} \delta_{x_{n,j}^P}}{\sum_{j=1}^J g_{n,j}} \approx \frac{1}{J} \sum_{j=1}^{J} \delta_{x_{n,j}^F}. \tag{S68}$$

In a slight abuse of notation, we will identify $x_{n,1:J}^P \equiv \eta_n^J$, and $x_{n,1:J}^F \equiv \pi_n^J$. As in Karjalainen et al. (2023), one can view this as an inhomogenous Markov process evolving on $\mathcal{X}^J$. The corresponding Markov transition kernel is then

$$\mathbf{M}_n(x_{n-1,1:J}^P, \cdot) = \left(\Phi_n\left(\eta_{n-1}^J\right)\right)^{\otimes J} = \left(\Phi_n\left(\frac{1}{J}\sum_{j=1}^{J} \delta_{x_{n-1,j}^P}\right)\right)^{\otimes J}, \tag{S69}$$

and the composition of Markov kernels on particles from timestep n to timestep $n+k$ is written

$$\mathbf{M}_{n,n+k} = \mathbf{M}_{n+1} \cdots \mathbf{M}_{n+k}. \tag{S70}$$

One may wonder why Karjalainen et al. (2023) require this process to evolve on $\mathcal{X}^J$. This is because at every timestep n, we in fact draw $X_{n,j}^P | \{X_{n-1,1:J}^P = x_{n-1,1:J}^P\} \sim \mathbf{M}_n(x_{n-1,1:J}^P, \cdot) = \eta_0^{\otimes J} \mathbf{M}_{0,n}$ for $j = 1, ..., J$.

Forgetting of the particle filter: The above formalization yields a result from Karjalainen et al. (2023) on the forgetting of the particle filter that we require for our analysis of the bias, variance, and error of MOP-α. That is, Karjalainen et al. (2023) show that

$$\beta_{\mathrm{TV}}\left(\mathbf{M}_{n,n+k}\right) \leq (1-\epsilon)^{\lfloor k/(O(\log J)) \rfloor}, \tag{S71}$$

for some ϵ dependent on $\bar{G}, \underline{\mathrm{G}}, \bar{M}, \underline{\mathrm{M}}$ in Assumptions (A3) and (A2). As a result, the mixing time of the particle filter is only on the order of $O(\log(J))$ timesteps.

Equipped with the above formalisms and results, we are now in a position to provide guarantees on the performance of MOP-α itself.

S4 Proof of Theorem 3 via a Strong Law of Large Numbers for Triangular Arrays of Particles With Off-Parameter Resampling

Here, we prove Theorem 3 by deriving more general result, from which it will be clear that MOP-α targets the filtering distribution and is strongly consistent for the likelihood. We will prove a strong law of large numbers for triangular arrays of particles with off-parameter resampling, meaning that we resample the particles according to an arbitrary resampling rule that is not necessarily in proportion to the target distribution of interest. Using weights that encode the cumulative discrepancy between the resampling distribution and the target distribution (instead of resampling to equal weights, as in the basic particle filter) provides a sufficient correction to ensure almost sure convergence. We now introduce the precise definition of an off-parameter resampled particle filter.

Definition S1 (Off-Parameter Resampled Particle Filters)**.** *An off-parameter resampled particle filter is a Monte Carlo approximation to a Feynman-Kac model* $\left(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N\right)$*, where we inductively define given some* $x^F_{n-1,j}$*,* $w^F_{n-1,j}$ *comprising the filtering measure approximation* π^J_n*:*

1. *The prediction particles at timestep* n*,* $x^P_{n,j}$*, are drawn from* $X^P_{n,j} \sim \pi^J_n M_n$*, and the prediction weights are given by* $w^P_{n,j} = w^F_{n-1,j}$*.*
2. *The prediction measure at timestep* n *is given by*

$$\eta^J_n = \frac{\sum_{j=1}^J w^P_{n,j}\, \delta_{x^P_{n,j}}}{\sum_{j=1}^J w^P_{n,j}}. \tag{S72}$$

3. *The particles are resampled at every timestep* n*, yielding indices* k_j *according to some arbitrary probabilities* $(p_{n,j})_{j=1}^J$*.*
4. *The filtering measure at timestep* n *is given by*

$$\pi^J_n = \frac{\sum_{j=1}^J \delta_{x^P_{n,j}}\, w^P_{n,j}\, g_{n,j}}{\sum_{j=1}^J w^P_{n,j}\, g_{n,j}} \approx \frac{\sum_{j=1}^J \delta_{x^P_{n,k_j}}\, w^P_{n,k_j}\, g_{n,k_j}/p_{n,k_j}}{\sum_{j=1}^J w^P_{n,k_j}\, g_{n,k_j}/p_{n,k_j}} = \frac{\sum_{j=1}^J w^F_{n,j}\, \delta_{x^F_{n,j}}}{\sum_{j=1}^J w^F_{n,j}}. \tag{S73}$$

5. *The prediction weights at the next timestep are given by* $w^P_{n+1,j} = w^F_{n,j} = w^P_{n,k_j}\, g_{n,k_j}/p_{n,k_j}$*.*

To prove that the weight correction is sufficient for almost sure convergence, we first introduce what it means for a triangular array of particles to *target* a given target distribution.

Definition S2 (Targeting)**.** *A pair of random vectors* (X, W) *drawn from some measure* g ***targets*** *another measure* π *if, for any measurable and bounded functional* h*,*

$$E_g\big[h(X)\cdot W\big] = E_\pi\big[h(X)\big]. \tag{S74}$$

A particle approximation is a triangular array of pairs of random vectors (X^J_j, W^J_j)*,* $j = 1, 2, \ldots, J$*. The particle approximation* ***targets*** π *if, for any measurable and bounded functional* h*,*

$$\frac{\sum_{j=1}^J h(X^J_j)\, W^J_j}{\sum_{j=1}^J W^J_j} \overset{a.s.}{\to} E_\pi\big[h(X)\big] \tag{S75}$$

as $J \to \infty$*.*

Chopin (2004) asserted without proof that common particle filter algorithms target the filtering distribution in this sense, while Chopin and Papaspiliopoulos (2020) proved a related result assuming bounded densities. We follow a similar approach to Chopin and Papaspiliopoulos (2020), based on showing strong laws of large numbers for triangular arrays, noting that triangular array strong laws do not generally hold without an additional regularity condition such as boundedness.

In order to prove the consistency of our variation on the particle filter, we now present three helper lemmas. The first follows from standard importance sampling arguments, the second from integrating out the marginal, and the third from Bayes' theorem. We state Lemma S5 assuming multinomial resampling, which is convenient for the proof though other resampling strategies may be preferable in practice.

Lemma S5 (Change of Weight Measure)**.** *Suppose that* $\{(\tilde{X}_j^J, U_j^J), j = 1, \ldots, J\}$ *targets* f_X*. Now, let* $\{(Y_j^J, V_j^J), j = 1, \ldots, J\}$ *be a multinomial sample with indices* k_j *drawn from* $\{(\tilde{X}_j^J, U_j^J)\}$ *where* $(\tilde{X}_j^J, U_j^J)$ *is represented, on average, proportional to* $\pi_j^J J$ *times. Write*

$$(Y_j^J, V_j^J) = \big(\tilde{X}_{k_j}^J, U_{k_j}^J / \pi_{k_j}^J\big). \tag{S76}$$

If the importance sampling weights U_j/π_j *are bounded, then* $\{(Y_j^J, V_j^J), j = 1, \ldots, J\}$ *targets* f_X*.*

Proof. Note that as the Y_j^J are a subsample from X_j^J, h can be a function of Y as well as it is one for X. We then expand

$$\frac{\sum_j h(Y_j^J)\, V_j^J}{\sum_j V_j^J} = \frac{\sum_j h(\tilde{X}_{k_j}^J)\, \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}}. \tag{S77}$$

By hypothesis,

$$\frac{\sum_j h(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[f(X)\big]. \tag{S78}$$

We want to show

$$\frac{\sum_j h(X_{k_j}^J)\, \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}} - \frac{\sum_j h(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} 0. \tag{S79}$$

For this, it is sufficient to show that

$$\sum_j h(\tilde{X}_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(\tilde{X}_j^J) \frac{U_j^J}{\pi_j^J} \pi_j^J \overset{a.s.}{\to} 0 \tag{S80}$$

since an application of this result with $h(x) = 1$ provides almost sure convergence of the denominator in (S79). Write $g(\tilde{X}_j^J) = h(\tilde{X}_j^J) \frac{U_{k_j}^J}{\pi_{k_j}^J}$. We therefore need to show that

$$\sum_j Z_j^J := \sum_j \Big(g(\tilde{X}_{k_j}^J) - g(\tilde{X}_j^J)\, \pi_j^J \Big) \overset{a.s.}{\to} 0. \tag{S81}$$

Because the functional h and importance sampling weights $u^J_{k_j}/\pi^J_{k_j}$ are bounded, we have that $\mathbb{E}\big[(Z^J_j)^4\big] < \infty$. We can then follow the argument of Chopin and Papaspiliopoulos (2020) from this point on, where noting that the Z^J_j are conditionally independent given the $\tilde{X}^J_j$ and U^J_j,

$$
\begin{aligned}
\mathbb{E}\left[\left(\sum_j Z^J_j\right)^4 \middle| (\tilde{X}^J_j, U^J_j)^J_{j=1}\right] &= J\,\mathbb{E}\left[(Z^J_1)^4 | (\tilde{X}^J_j, U^J_j)^J_{j=1}\right] \\
&\quad + 3J(J-1)\left(\mathbb{E}\big[(Z^J_j)^2 \big| (\tilde{X}^J_j, U^J_j)^J_{j=1}\big]\right) && \text{(S82)} \\
&\leq CJ^2, && \text{(S83)}
\end{aligned}
$$

for some $C > 0$. Taking expectations on both sides yields

$$
\mathbb{E}\left[\left(\sum_j Z^J_j\right)^4\right] \leq CJ^2 \tag{S84}
$$

by the tower property. Now by Markov's inequality,

$$
\mathbb{P}\left(\left|\frac{1}{J}\sum_{j=1}^J Z^J_j\right| > \epsilon\right) \leq \frac{1}{\epsilon^4 J^4}\mathbb{E}\left[\left(\sum_{j=1}^J Z^J_j\right)^4\right] \leq \frac{C}{\epsilon^4 J^2}, \tag{S85}
$$

and as these terms are summable we can apply Borel-Cantelli to conclude that these deviations happen only finitely often for every $\epsilon > 0$, giving us the almost-sure convergence for

$$
\sum_j h(X^J_{k_j})\,\frac{u^J_{k_j}}{\pi^J_{k_j}} - \sum_j h(X^J_j)\,\frac{u^J_j}{\pi^J_j}\,\pi^J_j \overset{a.s.}{\to} 0. \tag{S86}
$$

Similarly, we also have that

$$
\sum_j \frac{u^J_j}{\pi^J_j}\pi^J_j - \sum_j \frac{u^J_{k_j}}{\pi^J_{k_j}} \overset{a.s.}{\to} 0, \tag{S87}
$$

and the result is proved.

□

Remark: Note that Lemma S5 permits $\pi_{1:J}$ to depend on $\{(X_j, u_j)\}$ as long as the resampling is carried out independently of $\{(X_j, u_j)\}$, conditional on $\pi_{1:J}$.

Lemma S6 (Particle Marginals)**.** *Suppose that $\{(\tilde{X}^J_j, U^J_j), j = 1, \ldots, J\}$ targets f_X. Also suppose that $\tilde{Z}^J_j \sim f_{Z|X}(\cdot|\tilde{X}^J_j)$ where $f_{Z|X}$ is a conditional probability density function corresponding to a joint density $f_{X,Z}$ with marginal densities f_X and f_Z. Then, if the U^J_j are bounded, $\{(\tilde{Z}^J_j, U^J_j)\}$ targets f_Z.*

Proof. We want to show that, for any measurable bounded h,

$$
\frac{\sum_j h(\tilde{Z}^J_j)\,U^J_j}{\sum_j U^J_j} \overset{a.s.}{\to} \mathbb{E}_{f_Z}[h(Z)] = \mathbb{E}_{f_X}\left[\mathbb{E}_{f_{Z|X}}[h(Z)|X]\right]. \tag{S88}
$$

By assumption, for any measurable and bounded functional g with domain $\mathcal{X}$,

$$\frac{\sum_j g(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}[g(X)]. \tag{S89}$$

Let $\bar{U}_j^J = \frac{J\, U_j^J}{\sum_j U_j^J}$. Examine the numerator and denominator of the quantity

$$\frac{J^{-1}\sum_j h(\tilde{Z}_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} = \frac{J^{-1}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J}{J^{-1}\sum_j \bar{U}_j^J}. \tag{S90}$$

The denominator converges to 1 almost surely. The numerator, on the other hand, is

$$\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J, \tag{S91}$$

and by the same fourth moment argument to the above lemma, it converges almost surely to the limit of its expectation,

$$\begin{aligned} \lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J\right] &= \lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j \mathbb{E}\Big[h(\tilde{Z}_j^J)\, \bar{U}_j^J \Big| \tilde{X}_j^J, \bar{U}_j^J\Big]\right] && \text{(S92)} \\ &= \lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j \mathbb{E}\Big[h(Z)\Big| X = \tilde{X}_j^J\Big]\, \bar{U}_j^J\right]. && \text{(S93)} \end{aligned}$$

Applying (S89) with $g(x) = \mathbb{E}\big[h(Z)|X = x\big]$, the average on the right hand side converges almost surely to $\mathbb{E}\big\{\mathbb{E}[h(Z)|X]\big\} = \mathbb{E}[h(Z)]$. It remains to swap the limit and expectations. We can do so with the bounded convergence theorem, and therefore obtain

$$\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J \overset{a.s.}{\to} \mathbb{E}_{f_Z}[h(Z)]. \tag{S94}$$

□

Lemma S7 (Particle Posteriors)**.** *Suppose that* $\{(X_j^J, U_j^J), j = 1, \ldots, J\}$ *targets* f_X. *Also suppose that* $(X_j'^J, U_j'^J) = \big(X_j^J, U_j^J\, f_{Z|X}(z^*|X_j^J)\big)$. *Then, if* $U_j^J\, f_{Z|X}(z^*|X_j^J)$ *and* $U_j^J\, f_{Z|X}(z^*|X_j^J)\big/f_Z(z^*)$ *are bounded,* $\{(X_j'^J, U_j'^J)\}$ *targets* $f_{X|Z}(\cdot|z^*)$.

Proof. Again, we want to show that

$$\frac{\sum_j h(X_j^J)\cdot U_j^J \cdot f_{Z|X}(z^*|X_j^J)}{\sum_j U_j^J \cdot f_{Z|X}(z^*|X_j^J)} \overset{a.s.}{\to} \mathbb{E}_{f_{X|Z}}\big[h(X)\big|z^*\big]. \tag{S95}$$

We already have that for any measurable bounded g,

$$\frac{\sum_j g(X_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[g(X)\big]. \tag{S96}$$

Consider the following:

$$\frac{J^{-1}\sum_j h(X_j^J)\, f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} \times \left(\frac{J^{-1}\sum_j f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J}\right)^{-1}. \tag{S97}$$

We will apply Equation (S96) to the numerator and the denominator in the ratio above individually. The numerator converges to

$$\frac{J^{-1}\sum_j h(X_j^J)\, f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j u_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[h(X)\, f_{Z|X}(z^*|X)\big], \tag{S98}$$

while the reciprocal of the denominator converges to

$$\frac{J^{-1}\sum_j f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[f_{Z|X}(z^*|X)\big] = f_Z(z^*). \tag{S99}$$

Now we take advantage of the identities

$$\frac{\mathbb{E}_{f_X}\big[h(X)\, f_{Z|X}(z^*|X)\big]}{f_Z(z^*)} = \mathbb{E}_{f_X}\left[\frac{h(X)\, f_{Z|X}(z^*|X)}{f_Z(z^*)}\right] \tag{S100}$$

$$= \mathbb{E}_{f_X}\left[h(X)\frac{f_{X|Z}(X|z^*)}{f_X(X)}\right] = \mathbb{E}_{f_{X|Z}}\big[h(X)|z^*\big], \tag{S101}$$

to give the desired result.

□

Theorem S1 (Off-Parameter Resampled Particle Filters Target the Filtering Distribution)**.** *The off-parameter resampled particle filter as outlined in Definition S1 targets the filtering distribution.*

Proof. Recursively applying Lemmas S5, S6, and S7, we obtain that the off-parameter resampled particle filter targets the posterior. Specifically, suppose inductively that $\big\{\big(X_{n-1,j}^F, w_{n-1,j}^F\big)\big\}$ targets π_{n-1}. Then, Lemma S6 tells us that $\big\{\big(X_{n,j}^P, w_{n,j}^P\big)\big\}$ targets η_n. Lemma S7 tells us that $\big\{\big(X_{n,j}^P, w_{n,j}^P g_{n,j}^\theta\big)\big\}$ therefore targets π_n. Lemma S5 guarantees that the resampling rule, given by

$$\big(X_{n,j}^F, w_{n,j}^F\big) = \big(X_{n,k_j}^P, w_{n,k_j}^P\, g_{n,k_j}/p_{n,k_j}\big), \tag{S102}$$

with resampling probabilities proportional to $p_{n,j}$, therefore also targets π_n. □

Proposition S1 (MOP-1 Targets the Filtering Distribution)**.** *When $\alpha = 1$ or $\phi = \theta$, MOP-α targets the filtering distribution.*

Proof. When $\theta = \phi$, regardless of the value of α, the ratio $\frac{g_{n,j}^\theta}{g_{n,j}^\phi} = 1$, and this reduces to the vanilla particle filter estimate.

When $\alpha = 1$, and $\theta \neq \phi$, the proof is identical to that of S1. Recursively applying Lemmas S5, S6, and S7, we obtain that the MOP-1 filter targets the posterior. Specifically, suppose inductively that $\big\{\big(X_{n-1,j}^{F,\theta}, w_{n-1,j}^{F,\theta}\big)\big\}$ is properly weighted for $f_{X_{n-1}|Y_{1:n-1}}(x_{n-1}|y_{1:n-1}^*;\theta)$. Then, Lemma S6 tells us that $\{(X^{P,\theta}_{n,j}, w^{P,\theta}_{n,j})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y^*_{1:n-1};\theta)$. Lemma S7 tells us that $\{(X^{P,\theta}_{n,j}, w^{P,\theta}_{n,j}\, g^{\theta}_{n,j})\}$ therefore targets $f_{X_n|Y_{1:n}}(x_n|y^*_{1:n};\theta)$. Lemma S5 guarantees that the resampling rule, given by

$$\big(X^{F,\theta}_{n,j}, w^{F,\theta}_{n,j}\big) = \big(X^{P,\theta}_{n,k_j}, w^{P,\theta}_{n,k_j}\, g^{\theta}_{n,k_j}/g^{\phi}_{n,k_j}\big), \tag{S103}$$

with resampling weights proportional to $g^{\phi}_{n,j}$, therefore also targets $f_{X_n|Y_{1:n}}(x_n|y^*_{1:n};\theta)$. □

This has addressed filtering, but not quite yet the likelihood evaluation. For this we use the following lemma.

Lemma S8 (Likelihood Proper Weighting)**.** $f_{Y_n|Y_{1:n-1}}(y^*_n|y^*_{1_n-1};\theta)$ *is strongly consistently estimated by either the before-resampling estimate,*

$$L^{B,\theta}_n = \frac{\sum_{j=1}^J g^{\theta}_{n,j}\, w^{P,\theta}_{n,j}}{\sum_{j=1}^J w^{P,\theta}_{n,j}}, \tag{S104}$$

or by the after-resampling estimate,

$$L^{A,\theta}_n = L^{\phi}_n \frac{\sum_{j=1}^J w^{F,\theta}_{n,j}}{\sum_{j=1}^J w^{P,\theta}_{n,j}}. \tag{S105}$$

where L^{ϕ}_n is as defined in the various algorithms.

Here, (S104) is a direct consequence of our earlier result that $\{(X^{P,\theta}_{n,j}, w^{P,\theta}_{n,j})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y^*_{1:n-1};\theta)$. To see (S105), we write the numerator of (S5) as

$$L^{\phi}_n \sum_{j=1}^J \left[\frac{g^{\theta}_{n,j}}{g^{\phi}_{n,j}}\, w^{P,\theta}_{n,j}\right] \frac{g^{\phi}_{n,j}}{L^{\phi}_n} = L^{\phi}_n \sum_{j=1}^J w^{FC,\theta}_{n,j}\, \frac{g^{\phi}_{n,j}}{L^{\phi}_n}. \tag{S106}$$

Using Lemma S5, we resample according to probabilities $\frac{g^{\phi}_{n,j}}{L^{\phi}_n}$ to see this is properly estimated by

$$L^{\phi}_n \sum_{j=1}^J w^{F,\theta}_{n,j}, \tag{S107}$$

from which we obtain (S105). Using Lemma S8, we obtain a likelihood estimate,

$$L^{A,\theta} = \prod_{n=1}^N \left(L^{\phi}_n \frac{\sum_{j=1}^J w^{F,\theta}_{n,j}}{\sum_{j=1}^J w^{P,\theta}_{n,j}} \right). \tag{S108}$$

Since $w^{F,\theta}_{n,j} = w^{P,\theta}_{n+1,j}$, this is a telescoping product. The remaining terms are $\sum_{j=1}^J w^{P,\theta}_{0,j} = J$ on the denominator and $\sum_{j=1}^J w^{F,\theta}_{N,j}$ on the numerator. This derives the MOP-1 likelihood estimates.

$L^{B,\theta}$ should generally be preferred in practice, since there is no reason to include the extra variability from resampling when calculating the conditional log likelihood, but it lacks the nice telescoping product that lets us derive exact expressions for the gradient in Theorem 1.

S5 Proof of Theorem 4 via Consistency of Off-Parameter Resampled Gradient Estimates

We now provide a result showing that under sufficient regularity conditions, one can interchange the order of differentiation and expectation of functionals of off-parameter resampled particle estimates, showing that if an off-parameter resampled particle estimate is consistent for some estimand, its derivative is consistent for the derivative of the estimand as well.

One may wonder why this may be necessary, given that we have already shown strong consistency for measurable bounded functionals in Theorem S1. If we require the gradient to be bounded as well, then the gradient is also a bounded functional. The answer is that the strong consistency is for expectations under the filtering distribution, so Theorem S1 only establishes strong consistency of the gradient of the estimate to its expectation under the filtering distribution, $\nabla_\theta \eta_n^J(h_\theta) \overset{a.s.}{\to} \eta_n(\nabla_\theta h_\theta)$, which is not a-priori the gradient of the estimand $\nabla_\theta \eta_n(h_\theta)$. We require an interchange of the derivative and expectation, which we show is possible when the particle estimates have two continuous uniformly bounded derivatives over all J below.

Theorem S2 (Off-Parameter Particle Filters Yield Strongly Consistent Estimates of Derivatives of Functionals)**.** *Let $h_\theta : \mathcal{X} \to \mathbb{R}$ be a measurable bounded functional of particles, where $\eta_n^J(h_\theta)$ has two continuous derivatives uniformly bounded over all J by H^* for almost every $\omega \in \Omega$ and $\theta \in \Theta$. If it holds that $\eta_n^J(h_\theta) \overset{a.s.}{\to} \eta(h_\theta) = h_\theta^*$, then we also have that $\nabla_\theta \eta_n^J(h_\theta) \overset{a.s.}{\to} \eta_n(\nabla_\theta h_\theta) = \nabla_\theta \eta_n(h_\theta) = \nabla_\theta h_\theta^*$.*

Proof. Fix $\omega \in \Omega$. The sequence $(\nabla_\theta \eta_n^J(h_\theta)(\omega))_{J\in\mathbb{N}}$ is uniformly bounded over all J, by assumption. The sequence is also uniformly equicontinuous. To see this, by assumption, the second derivative of $\eta_n^J(h_\theta)(\omega)|_{\theta=\theta'}$ is also uniformly bounded over all J for almost every $\omega \in \Omega$ and every $\theta' \in \Theta$. A set of functions with derivatives bounded by the same constant is uniformly Lipschitz, and therefore uniformly equicontinuous. So the sequence $(\eta_n^J(h_\theta)\omega))_{J\in\mathbb{N}}$ is uniformly equicontinuous over θ for almost every $\omega \in \Omega$. Explicitly, for almost every $\omega \in \Omega$ and every $\epsilon > 0$, there exists some $\delta(\omega) > 0$ such that for every $||\theta - \theta'||_\infty < \delta$ and every $J \in \mathbb{N}$ we have that

$$\big|\big|\eta_n^J(h_\theta)(\omega) - \eta_n^J(h_{\theta'})(\omega)\big|\big|_\infty < \epsilon. \tag{S109}$$

Then, by Arzela-Ascoli, there exists a uniformly convergent subsequence. We claim that there is only one subsequential limit. When the gradient is bounded, we can treat the gradient as a bounded functional. So by Theorem S1 the sequence $(\eta_n^J(h_\theta)(\omega))_{J\in\mathbb{N}}$ converges pointwise for $\theta = \phi$ and almost every $\omega \in \Omega$, and there is therefore only one subsequential limit. The sequence therefore converges uniformly to its limit $\lim_{J\to\infty} \eta_n^J(h_\theta)(\omega)$. Therefore, with uniform convergence for the derivatives established, we can swap the limit and derivative, and obtain that for almost every $\omega \in \Omega$,

$$\lim_{J\to\infty} \eta_n^J(h_\theta)(\omega) = \nabla_\theta \lim_{J\to\infty} \eta_n^J(h_\theta)(\omega). \tag{S110}$$

Again from Theorem S1, we know that

$$\eta_n^J(h_\theta)(\omega) \to \eta_n(h_\theta) = h_\theta^* \tag{S111}$$

for almost every $\omega \in \Omega$. We then have that for almost every $\omega \in \Omega$,

$$\lim_{J\to\infty} \eta_n^J(h_\theta)(\omega) = \nabla_\theta \eta_n(h_\theta) = \nabla_\theta h_\theta^*, \tag{S112}$$

as we wanted. □

Theorem 4 is proved as a corollary of Theorem S2, where we apply $\eta_n^J(h_\theta) = \hat{\mathcal{L}}_J^1(\theta)$, $\eta_n(h_\theta) = \mathcal{L}(\theta) = h_\theta^*$, and then use the continuous mapping theorem.

S6 Proof of Theorem 5 (Bias-Variance Analysis)

In this section, we prove various rates on the bias, variance, and MSE of DMOP-α. The earlier truncation argument is retained below for comparison. The current DMOP variables are the particle location and the accumulated weight gradient. Forgetting of the location alone does not immediately control the accumulated genealogical score. We prove the DMOP-0 result below and retain the former extension to $\alpha > 0$ inactive for comparison. Section S7 gives an exact finite-J genealogy audit, the bounds that remain valid for every α, and the additional centered estimate needed for a sharper variance theorem.

First, we note the following relation between the Dobrushin contraction coefficient and the alpha-mixing coefficients in our context below.

Lemma S9. *Setting X to be the particle collection at time m, and Y at time n, we have that the alpha mixing coefficients,*

$$\int \left| f_{XY}(x,y) - f_X(x)\, f_Y(y) \right| dx\, dy < \alpha, \tag{S113}$$

are bounded by the Dobrushin coefficient, i.e we have

$$\int \left| f_{Y|X}(y|x_1) - f_{Y|X}(y|x_2) \right| dy < \alpha \tag{S114}$$

for all x_1, x_2.

Proof. We rewrite the alpha-mixing assertion as

$$\int \int \left| f_{Y|X}(y|x) - f_Y(y) \right| dy\, f_X(x)\, dx. \tag{S115}$$

We claim that the Dobrushin coefficient implies

$$\int \left| f_{Y|X}(y|x_1) - f_Y(y) \right| dy < \alpha \tag{S116}$$

for all x_1. This is shown as follows:

$$\begin{aligned} \int \left| f_{Y|X}(y|x) - f_Y(y) \right| dy &= \int \left| \int \left[f_{Y|X}(y|x_1) - f_{Y|X}(y|x) \right] f_X(x)\, dx \right| dy && \text{(S117)} \\ &< \int \int \left| f_{Y|X}(y|x_1) - f_{Y|X}(y|x) \right| dy\, f_X(x)\, dx && \text{(S118)} \\ &< \int \alpha f_X(x)\, dx && \text{(S119)} \\ &= \alpha && \text{(S120)} \end{aligned}$$

We then have the desired result. □

As a warm-up, we consider a variance bound for DMOP-0. Note that we made Assumptions (A2) and (A3) to leverage the results from Karjalainen et al. (2023) on the forgetting of the particle filter. This is used below to recover the $O(N/J)$ variance bound for DMOP-0. The former truncation extension to $\alpha > 0$ is retained inactive for comparison. Section S7 gives the exact genealogical audit showing why the current-filter mixing argument cannot be transferred to the accumulated score functional. We start by decomposing the MOP-α estimator as follows, for the case $\theta = \phi$,

$$\hat{\mathcal{L}}(\theta) := \prod_{n=1}^{N} L_n^{A,\theta,\alpha} = \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta}} = \prod_{n=1}^{N} L_n^{A,\theta,\alpha} = \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} (w_{n-1,j}^{F,\theta})^{\alpha}}. \tag{S121}$$

Now, to illustrate the proof strategy for the general case in an easier context, we first analyze the special case when $\alpha = 0$. We now prove the variance bound for this case, presented below. To our knowledge, this result for the variance of the gradient estimate of Naesseth et al. (2018) is novel.

Theorem S3 (Variance of MOP-0 Gradient Estimate)**.** *The variance of DMOP-*0*, i.e. the algorithm of Naesseth et al. (2018), is*

$$\mathrm{Var}\big(\nabla_\theta \hat{\ell}^0(\theta)\big) \lesssim \frac{Np\, G'(\theta)^2}{J}. \tag{S122}$$

Proof. When $\alpha = 0$, we have

$$\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^{N} L_n^{A,\theta,0} = \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta,0}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta,0}} \tag{S123}$$

$$= \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{1}{J}\sum_{j=1}^{J} s_{n,j} = \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{1}{J}\sum_{j=1}^{J} \frac{f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{P,\theta};\theta\big)}{f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{P,\phi};\phi\big)}. \tag{S124}$$

Similarly to the proof of Lemma S2, we define

$$s_{n,j} = \frac{f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{P,\theta};\theta\big)}{f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{P,\phi};\phi\big)}, \tag{S125}$$

and from the exact same arguments conclude that its gradient when $\theta = \phi$ is therefore

$$\nabla_\theta \hat{\ell}^0(\theta) := \sum_{n=1}^{N} \nabla_\theta \log\left(L_n^{\phi} \frac{1}{J}\sum_{j=1}^{J} s_{n,j}\right) \tag{S126}$$

$$= \sum_{n=1}^{N} \frac{\nabla_\theta \left(L_n^{\phi}\frac{1}{J}\sum_{j=1}^{J} s_{n,j}\right)}{\left(L_n^{\phi}\frac{1}{J}\sum_{j=1}^{J} s_{n,j}\right)} \tag{S127}$$

$$= \sum_{n=1}^{N} \frac{\sum_{j=1}^{J} \nabla_\theta s_{n,j}}{\sum_{j=1}^{J} s_{n,j}} \tag{S128}$$

$$= \sum_{n=1}^{N} \frac{1}{J}\sum_{j=1}^{J} \frac{\nabla_\theta f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{F,\theta};\theta\big)}{f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{F,\phi};\phi\big)} \tag{S129}$$

$$= \frac{1}{J} \sum_{n=1}^{N} \sum_{j=1}^{J} \nabla_\theta \log\left(f_{Y_n|X_n}\left(y_n^*|x_{n,j}^{F,\theta};\theta\right)\right), \tag{S130}$$

where we use the derivative of the logarithm, observe that $\sum_{j=1}^{J} s_{n,j} = J$ when $\theta = \phi$, and use the derivative of the logarithm where $\theta = \phi$ again. We do this to establish that the gradient of the log-likelihood estimate is given by the sum of terms over all N and J:

$$\nabla_\theta \hat{\ell}^0(\theta) = \frac{1}{J} \sum_{n=1}^{N} \sum_{j=1}^{J} \nabla_\theta \log\left(f_{Y_n|X_n}\left(y_n^*|x_{n,j}^{F,\theta};\theta\right)\right). \tag{S131}$$

Therefore, for a given θ_i in the parameter vector θ,

$$\operatorname{Var}\left(\frac{\partial}{\partial\theta_i}\hat{\ell}^0(\theta)\right) = \frac{1}{J^2}\operatorname{Var}\left(\sum_{n=1}^{N}\sum_{j=1}^{J}\frac{\partial}{\partial\theta_i}\log\left(f_{Y_n|X_n}\left(y_n^*|x_{n,j}^{F,\theta};\theta\right)\right)\right) \tag{S132}$$

$$= \frac{1}{J^2}\sum_{n=1}^{N}\operatorname{Var}\left(\sum_{j=1}^{J}\frac{\partial}{\partial\theta_i}\log\left(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta};\theta)\right)\right)$$

$$+ 2\sum_{m<n}\operatorname{Cov}\left(\frac{1}{J}\sum_{j=1}^{J}\frac{\partial}{\partial\theta_i}\log\left(f_{Y_m|X_m}(y_m^*|x_{m,j}^{F,\theta};\theta)\right), \frac{1}{J}\sum_{j=1}^{J}\frac{\partial}{\partial\theta_i}\log\left(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta};\theta)\right)\right)$$

$$= \sum_{n=1}^{N}\operatorname{Var}\left(\frac{\partial}{\partial\theta_i}\hat{\ell}_n^0(\theta)\right) + 2\sum_{m<n}\operatorname{Cov}\left(\frac{\partial}{\partial\theta_i}\hat{\ell}_m^0(\theta), \frac{\partial}{\partial\theta_i}\hat{\ell}_n^0(\theta)\right). \tag{S133}$$

Here, we use Assumptions (A2) and (A3) that ensure strong mixing. We know from Theorem 3.1 of Karjalainen et al. (2023) that when $\mathbf{M}_{n,n+k}$ is the k-step Markov operator from timestep n and $\beta_{\mathrm{TV}}(M) = \sup_{x,y\in E}\|M(x,\cdot)-M(y,\cdot)\|_{\mathrm{TV}} = \sup_{\mu,\nu\in\mathcal{P},\mu\neq\nu}\frac{\|\mu M-\nu M\|_{\mathrm{TV}}}{\|\mu-\nu\|_{\mathrm{TV}}}$ is the Dobrushin contraction coefficient of a Markov operator,

$$\beta_{\mathrm{TV}}(\mathbf{M}_{n,n+k}) \leq (1-\epsilon)^{\lfloor k/(c\log(J))\rfloor}, \tag{S134}$$

i.e. the mixing time of the particle filter is $O(\log(J))$, where ϵ and c depend on $\bar{M}, \underline{\mathrm{M}}, \bar{G}, \underline{\mathrm{G}}$ in (A2) and (A3).

Lemma 2.3 of Karjalainen et al. (2023), followed by conditional multinomial resampling and the bounded Bayes-ratio step, gives the centered L^4 bound below. Theorem 3.1 of the same paper gives the displayed particle-cloud mixing coefficient. At $\alpha = 0$ these two results give

$$\left\|\frac{\partial}{\partial\theta_i}\hat{\ell}_n^0(\theta) - \mathbb{E}\left[\frac{\partial}{\partial\theta_i}\hat{\ell}_n^0(\theta)\right]\right\|_{L^4} \leq \frac{CG'(\theta)}{\sqrt{J}},$$
$$a(k) \leq (1-\epsilon)^{\lfloor (k-1)/[c\{1+\log J\}]\rfloor}. \tag{S135}$$

Davydov's inequality must be applied to the centered variables. Since covariance is unchanged by centering, Equation (S135) gives

$$\operatorname{Cov}\left(\frac{\partial}{\partial\theta_i}\hat{\ell}_m^0(\theta), \frac{\partial}{\partial\theta_i}\hat{\ell}_n^0(\theta)\right) \leq (1-\epsilon)^{\frac{1}{2}\lfloor\frac{|n-m|-1}{c\{1+\log(J)\}}\rfloor}\frac{G'(\theta)^2}{J}. \tag{S136}$$

Putting it all together, we see that

$$\operatorname{Var}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0(\theta)\right) = \sum_{n=1}^{N}\operatorname{Var}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}_n^0(\theta)\right) + 2\sum_{m<n}\operatorname{Cov}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}_m^0(\theta), \frac{\partial}{\partial_{\theta_i}}\hat{\ell}_n^0(\theta)\right) \tag{S137}$$

$$\leq \frac{NG'(\theta)^2}{J} + 2\sum_{n=1}^{N-1}(N-n)(1-\epsilon)^{\frac{1}{2}\lfloor\frac{n-1}{c\{1+\log(J)\}}\rfloor}\frac{G'(\theta)^2}{J} \tag{S138}$$

$$\leq \frac{G'(\theta)^2}{J}\left(N + 2\sum_{n=1}^{N-1}(N-n)(1-\epsilon)^{\frac{1}{2}\lfloor\frac{n-1}{c\{1+\log(J)\}}\rfloor}\right) \tag{S139}$$

$$\lesssim \frac{G'(\theta)^2 N}{J}. \tag{S140}$$

It then follows that as $\theta \in \Theta \subseteq \mathbb{R}^p$,

$$\operatorname{Var}\big(\nabla_\theta\hat{\ell}^0(\theta)\big) \lesssim \frac{Np\,G'(\theta)^2}{J}. \tag{S141}$$

□

Note that the factor of N that pops up here is due to the use of the unnormalized gradient. If one divides the gradient estimate by $\sqrt{N}$ before usage, the variance does not depend on the horizon. If one divides by N, the error is $O(1/\sqrt{NJ})$.

The order $NpG'(\theta)^2/J$ is the verified DMOP-0 baseline. It is not asserted to be a lower bound for every model or every value of α.

Theorem S4 (MSE and variance bounds for DMOP-α)**.** *Suppose the observations are fixed, $J \geq 2$, $\alpha \in [0,1)$, $\theta = \phi$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. The bounded measurement-gradient input in Assumption (A3) is understood to be the current function $h_n(X_n)$ in Equation (S146). Then*

$$\operatorname{Var}\left\{\nabla_\theta\hat{\ell}^\alpha(\theta)\right\} \lesssim pG'(\theta)^2\left\{\frac{N}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2}\right\}, \tag{S142}$$

$$\mathbb{E}\left\|\nabla_\theta\hat{\ell}^\alpha(\theta) - \nabla_\theta\ell(\theta)\right\|_2^2 \lesssim pG'(\theta)^2\left[\frac{N}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2} + N^2\left\{\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}\right\}^2\right]. \tag{S143}$$

The population component of the fixed-observation bias satisfies

$$\|\nabla_\theta\ell^\alpha(\theta) - \nabla_\theta\ell(\theta)\|_2 \leq CN\sqrt{p}\,G'(\theta)\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}. \tag{S144}$$

If $J \geq N$, the second term in Equation (S142) is absorbed by the first, so

$$\operatorname{Var}\left\{\nabla_\theta\hat{\ell}^\alpha(\theta)\right\} \lesssim \frac{NpG'(\theta)^2}{J(1-\alpha)^2}. \tag{S145}$$

Section S7 gives the complete proof and an exact constant-weight calculation that checks the N and J dependence and explains the singularity as $\alpha \uparrow 1$.

Proof. Lemma S13 proves Equation (S142). Theorem S5 proves Equation (S143) and Equation (S144). □

S7 Sharper Bias-Variance Analysis

We now establish the estimates used in Theorem S4. Write $\nabla_\theta \ell_n^\alpha(\theta)$ for the nth conditional score produced by the deterministic differentiated population recursion. At $\theta = \phi$, its $\alpha = 1$ version is the exact conditional score $\nabla_\theta \ell_n(\theta)$. The current DMOP object needed below is the pair consisting of the particle location and the gradient of its log weight. The location follows the ordinary particle transition to which Assumption (A2) applies. Along each selected lineage, the second component obeys the existing recursion

$$\nabla_\theta \log w_{n,j}^{F,\theta}\Big|_{\theta=\phi} = \alpha\, \nabla_\theta \log w_{n-1,k_j}^{F,\theta}\Big|_{\theta=\phi} + h_n(x_{n,j}^{F,\phi}). \tag{S146}$$

Here h_n is the bounded current measurement-gradient function of the particle location. This is the derivative at $\theta = \phi$ of the log-space recursion

$$\begin{aligned}\log w_{n,j}^{P,\theta} &= \alpha \log w_{n-1,j}^{F,\theta},\\ \log w_{n,j}^{F,\theta} &= \alpha \log w_{n-1,k_j}^{F,\theta} + \log g_{n,k_j}^{\theta} - \log g_{n,k_j}^{\phi}.\end{aligned} \tag{S147}$$

At $\theta = \phi$, the log weights are zero but their gradients need not be. We do not assume that the location–log-weight-gradient pair is mixing in total variation. Instead, the proof expands the log-weight gradient into its single-time h_n contributions, applies location forgetting to those contributions, and sums their powers of α. Throughout this section, the observations are fixed. Expectations and variances are over the simulator and particle-filter random variables, conditional on those observations.

We use Assumptions (A3) and (A5) in the precise pathwise sense already required by the differentiated MOP construction. For fixed simulator random numbers and fixed resampling indices, Assumption (A5) says that the recursively simulated state path and the composed log measurement densities are continuously differentiable in θ near ϕ, and that the joint law of the base simulator random numbers does not depend on θ. The derivative in the last clause of Assumption (A3) is the resulting total derivative: uniformly in n, j and in a neighborhood of ϕ, after enlarging $G'(\theta)$ to bound that neighborhood if necessary,

$$\left\| \nabla_\theta \log g_{n,j}^{A,F,\theta} \right\|_\infty \le G'(\theta). \tag{S148}$$

These two existing assumptions also justify differentiation under the finite-horizon simulator expectations used below. Indeed, the product rule and Equation (S148) give, pathwise,

$$\begin{aligned}&\left\| \nabla_\theta \prod_{s=1}^t G_s(X_s) \right\|_\infty \\ &\quad= \left\| \prod_{s=1}^t G_s(X_s) \sum_{s=1}^t \nabla_\theta \log G_s(X_s) \right\|_\infty \\ &\le \prod_{s=1}^t G_s(X_s) \sum_{s=1}^t \| \nabla_\theta \log G_s(X_s) \|_\infty \\ &\le t \bar{G}^t G'(\theta).\end{aligned} \tag{S149}$$

The final expression is deterministic and finite for each fixed t. For each coordinate of θ, the mean-value theorem therefore bounds every difference quotient about ϕ by the same quantity $t\bar{G}^t G'(\theta)$. The dominated convergence theorem gives

$$\nabla_\theta \mathbb{E}_U \left[\prod_{s=1}^{t} G_s(X_s) \right] = \mathbb{E}_U \left[\nabla_\theta \prod_{s=1}^{t} G_s(X_s) \right]. \tag{S150}$$

Thus the required interchange follows from the clarified Assumptions (A3) and (A5); it is not an additional assumption. Geometric population forgetting follows from the backward-kernel contraction under Assumption (A2). Lemma S11 gives a uniform weak-error bound for a current filtering functional. Its proof is not used for a historical functional carried by the genealogy; that separate issue is audited below.

For the path functionals below, evaluated at $\theta = \phi$, let ρ_t be the posterior path measure of the particle locations. Thus, for every integrable path functional h,

$$\rho_t(h) = \frac{\mathbb{E}\left[h(X_{0:t}) \prod_{s=1}^{t} G_s(X_s) \right]}{\mathbb{E}\left[\prod_{s=1}^{t} G_s(X_s) \right]}. \tag{S151}$$

Here $G_s(X_s)$ is the existing measurement potential. The accumulated weight gradient is the second component of the current pair in Equation (S146); it is not included in the location variable X_s or in the transition density appearing in Assumption (A2).

Unrolling the existing filtered-weight recursion at $\theta = \phi$ gives

$$\nabla_\theta \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} = \sum_{r=0}^{n-1} \alpha^r \, \nabla_\theta \log g_{n-r,j}^{A,F,\theta} \Big|_{\theta=\phi}. \tag{S152}$$

For a complete ancestral path, write h_k, only in this section, for the measurement-score contribution

$$h_k = \nabla_\theta \log g_k^{A,F,\theta} \Big|_{\theta=\phi} \tag{S153}$$

carried on that path. In the path-space notation of Chan et al. (2018), the notation used below means explicitly

$$m(\xi_t) = \frac{1}{J} \sum_{j=1}^{J} \delta_{\xi_{t,j}}, \qquad m(\xi_t)(h_k) = \frac{1}{J} \sum_{j=1}^{J} h_k(\xi_{t,j}), \quad 1 \le k \le t, \tag{S154}$$

where $\xi_{t,j}$ is the complete ancestral particle path carried by particle j at time t. Thus $h_k(\xi_{t,j})$ is the single time-k measurement-score contribution on that particle, not an arbitrary functional denoted by m. Equation (S151) gives its population counterpart.

We now identify the lag coefficients in the conditional score. At $\theta = \phi$, all weights equal one, so differentiating the two sample averages in the definition of $L_n^{A,\theta,\alpha}$ gives

$$\begin{aligned} \nabla_\theta \hat{\ell}_n^\alpha(\theta) &= \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} - \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} \\ &= \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} - \frac{\alpha}{J} \sum_{j=1}^{J} \nabla_\theta \log w_{n-1,j}^{F,\theta} \Big|_{\theta=\phi}, \end{aligned} \tag{S155}$$

where the second equality uses $w_{n,j}^{P,\theta} = (w_{n-1,j}^{F,\theta})^{\alpha}$. In the MOP-$\alpha$ recursion, the discounted-weight accumulation and process propagation preserve the index j and occur before the time-n resampling step. The resampling index k_j enters only afterward, through

$$w_{n,j}^{F,\theta} = w_{n,k_j}^{P,\theta} \frac{g_{n,k_j}^{\theta}}{g_{n,k_j}^{\phi}}. \tag{S156}$$

Consequently, the second average in Equation (S155) is pathwise

$$\frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{P,\theta}\Big|_{\theta=\phi} = \frac{\alpha}{J}\sum_{j=1}^{J} \nabla_\theta \log w_{n-1,j}^{F,\theta}\Big|_{\theta=\phi}, \tag{S157}$$

which is the empirical average over the time-$(n-1)$ filtered genealogy. Substituting Equation (S152) into Equation (S155) and changing the index in the second sum give

$$\begin{aligned}
\nabla_\theta \hat{\ell}_n^{\alpha}(\theta) &= \sum_{r=0}^{n-1} \alpha^r m(\xi_n)(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r m(\xi_{n-1})(h_{n-r}) \\
&= m(\xi_n)(h_n) + \sum_{r=1}^{n-1} \alpha^r \left\{ m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r}) \right\}.
\end{aligned} \tag{S158}$$

Consequently, define

$$\begin{aligned}
\Delta_{n,0}^{J} &= m(\xi_n)(h_n), & \Delta_{n,0} &= \rho_n(h_n), \\
\Delta_{n,r}^{J} &= m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r}), & \Delta_{n,r} &= \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}), \quad 1 \le r < n.
\end{aligned} \tag{S159}$$

We now verify that the right-hand population quantities are the lag coefficients of the deterministic differentiated recursion. Equations (S150) and (S153) give, at $\theta = \phi$,

$$\begin{aligned}
&\nabla_\theta \log \mathbb{E}_U \left[\prod_{s=1}^{n} G_s(X_s) \right] \\
&\quad = \frac{\mathbb{E}_U \left[\prod_{s=1}^{n} G_s(X_s) \sum_{s=1}^{n} h_s \right]}{\mathbb{E}_U \left[\prod_{s=1}^{n} G_s(X_s) \right]} \\
&\quad = \sum_{s=1}^{n} \rho_n(h_s).
\end{aligned} \tag{S160}$$

Subtracting the same identity at time $n-1$ gives

$$\begin{aligned}
\nabla_\theta \ell_n(\theta) &= \sum_{s=1}^{n} \rho_n(h_s) - \sum_{s=1}^{n-1} \rho_{n-1}(h_s) \\
&= \rho_n(h_n) + \sum_{r=1}^{n-1} \left\{ \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}) \right\} \\
&= \sum_{r=0}^{n-1} \Delta_{n,r}.
\end{aligned} \tag{S161}$$

Finally, differentiating the two population weighted averages in the deterministic MOP-α recursion and using Equation (S150) gives

$$\begin{aligned}\nabla_\theta \ell_n^\alpha(\theta) &= \sum_{r=0}^{n-1} \alpha^r \rho_n(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r \rho_{n-1}(h_{n-r}) \\ &= \rho_n(h_n) + \sum_{r=1}^{n-1} \alpha^r \left\{\rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r})\right\} \\ &= \sum_{r=0}^{n-1} \alpha^r \Delta_{n,r}. \end{aligned} \tag{S162}$$

Lemma S10 (Lagwise first-derivative forgetting)**.** *Suppose $\theta = \phi$. Under Assumptions (A2), (A3), and (A5), there are constants $C < \infty$ and $\varrho \in (0,1)$ such that, for $0 \le r < n$,*

$$\|\Delta_{n,r}\|_2 \le C\sqrt{p}\, G'(\theta)\varrho^r. \tag{S163}$$

Proof. We derive the result from the posterior update and the contraction of the posterior backward kernels.

Step 1: one-step posterior update. First suppose that $1 \le r < n$, and put $k = n - r$. Fix a coordinate $i \in \{1, \dots, p\}$, and let $h_{k,i}$ be the corresponding coordinate of h_k in Equation (S153). By Equation (S148),

$$|h_{k,i}| \le G'(\theta). \tag{S164}$$

Because $h_{k,i}$ is measurable with respect to the canonical extended path through time $k \le n-1$, Equation (S151) and conditional expectation with respect to the new base simulator variable give

$$\begin{aligned}\rho_n(h_{k,i}) &= \frac{\mathbb{E}_U\left[h_{k,i}\prod_{s=1}^{n-1} G_s(X_s)\mathbb{E}_U\{G_n(X_n) \mid X_{n-1}\}\right]}{\mathbb{E}_U\left[\prod_{s=1}^{n-1} G_s(X_s)\mathbb{E}_U\{G_n(X_n) \mid X_{n-1}\}\right]} \\ &= \frac{\rho_{n-1}\{h_{k,i}M_n(G_n)\}}{\rho_{n-1}M_n(G_n)}, \end{aligned} \tag{S165}$$

where the first equality uses the independence of the new base simulator variable from the path through time $n-1$. The second equality first uses the identity below and then divides its numerator and denominator by $\mathbb{E}_U\{\prod_{s=1}^{n-1} G_s(X_s)\}$. Using the manuscript's kernel notation, the required identity is

$$M_n(G_n)(x_{n-1}) = \int M_n(x_{n-1}, dx_n)G_n(x_n). \tag{S166}$$

The bounds on G_n and the fact that M_n is a Markov kernel imply for every x_{n-1} that

$$M_n(G_n)(x_{n-1}) \ge \underline{\mathrm{G}} \int M_n(x_{n-1}, dx_n) = \underline{\mathrm{G}}, \tag{S167}$$

$$M_n(G_n)(x_{n-1}) \le \bar{G} \int M_n(x_{n-1}, dx_n) = \bar{G}. \tag{S168}$$

Step 2: backward-kernel contraction. We next control how much information about the path through time k can remain at time $n-1$. For $s \geq 1$, the posterior backward kernel from X_s to X_{s-1}, written only in this proof as B_s, is

$$B_s(x_s, dx_{s-1}) = \frac{\pi_{s-1}(dx_{s-1}) f_{X_s|X_{s-1}}(x_s \mid x_{s-1}; \theta)}{\displaystyle\int \pi_{s-1}(du) f_{X_s|X_{s-1}}(x_s \mid u; \theta)}. \tag{S169}$$

Assumption (A2) gives the following minorization for every measurable set A:

$$\begin{aligned} B_s(x_s, A) &= \frac{\displaystyle\int_A \pi_{s-1}(dx_{s-1}) f_{X_s|X_{s-1}}(x_s \mid x_{s-1}; \theta)}{\displaystyle\int \pi_{s-1}(du) f_{X_s|X_{s-1}}(x_s \mid u; \theta)} \\ &\geq \frac{\underline{\mathrm{M}}\, \pi_{s-1}(A)}{\bar{M} \int \pi_{s-1}(du)} \\ &= \frac{\underline{\mathrm{M}}}{\bar{M}} \pi_{s-1}(A). \end{aligned} \tag{S170}$$

Put $\epsilon = \underline{\mathrm{M}}/\bar{M}$. If $\epsilon < 1$, Equation (S170) permits the decomposition

$$B_s(x_s, \cdot) = \epsilon \pi_{s-1}(\cdot) + (1-\epsilon) R_s(x_s, \cdot), \tag{S171}$$

where

$$R_s(x_s, A) = \frac{B_s(x_s, A) - \epsilon \pi_{s-1}(A)}{1-\epsilon} \tag{S172}$$

is a Markov kernel. Hence, for every bounded real function f and any x_s, x_s',

$$\begin{aligned} B_s(f)(x_s) - B_s(f)(x_s') &= (1-\epsilon)\{R_s(f)(x_s) - R_s(f)(x_s')\}, \\ \left|B_s(f)(x_s) - B_s(f)(x_s')\right| &\leq (1-\epsilon) \operatorname{osc}(f). \end{aligned} \tag{S173}$$

If $\epsilon = 1$, Equation (S170) implies $B_s(x_s, \cdot) = \pi_{s-1}(\cdot)$, so the left-hand side is zero. Taking the supremum over x_s, x_s' gives

$$\operatorname{osc}\{B_s(f)\} \leq (1-\epsilon) \operatorname{osc}(f). \tag{S174}$$

Step 3: propagate the time-k contribution. The function $h_{k,i} = h_{k,i}(X_k)$ is a bounded current function of the particle location X_k. Repeated conditioning through the backward kernels therefore gives

$$\mathbb{E}_{\rho_{n-1}}\{h_{k,i}(X_k) \mid X_{n-1} = x_{n-1}\} = (B_{n-1} B_{n-2} \cdots B_{k+1} h_{k,i})(x_{n-1}). \tag{S175}$$

There are $n-1-k = r-1$ backward kernels in this display. Moreover, Equation (S164) gives

$$\begin{aligned} \operatorname{osc}(h_{k,i}) &\leq 2\|h_{k,i}\|_\infty \\ &\leq 2G'(\theta). \end{aligned} \tag{S176}$$

If $r = 1$, Equation (S175) contains no backward kernel. If $r \geq 2$, applying Equation (S174) successively gives

$$\operatorname{osc}\left[\mathbb{E}_{\rho_{n-1}}\{h_{k,i} \mid X_{n-1} = \cdot\}\right]$$

$$\leq 2G'(\theta)\begin{cases}1, & r=1,\\ (1-\epsilon)^{r-1}, & r\geq 2.\end{cases} \tag{S177}$$

Step 4: covariance identity and geometric bound. To finish, abbreviate the conditional expectation on the left of Equation (S177) by H, only for the next calculation. Since the X_{n-1}-marginal of ρ_{n-1} is π_{n-1}, the tower property gives

$$\rho_{n-1}(h_{k,i}) = \pi_{n-1}(H), \tag{S178}$$

$$\rho_{n-1}\{h_{k,i}M_n(G_n)\} = \pi_{n-1}\{HM_n(G_n)\}. \tag{S179}$$

Substituting Equations (S178)– (S179) into Equation (S165) and then subtracting $\rho_{n-1}(h_{k,i})$ gives the exact covariance identity

$$\begin{aligned}\Delta_{n,r,i} &= \rho_n(h_{k,i}) - \rho_{n-1}(h_{k,i})\\ &= \frac{\pi_{n-1}\{HM_n(G_n)\} - \pi_{n-1}(H)\pi_{n-1}M_n(G_n)}{\pi_{n-1}M_n(G_n)}\\ &= \frac{\pi_{n-1}[\{H-\pi_{n-1}(H)\}\{M_n(G_n)-\pi_{n-1}M_n(G_n)\}]}{\pi_{n-1}M_n(G_n)}.\end{aligned} \tag{S180}$$

For any probability measure μ and bounded real f,

$$|f(x)-\mu(f)| \leq \operatorname{osc}(f). \tag{S181}$$

The first positive lag also inherits one contraction from the forward transition. To see this directly, fix $x^\star$. Assumption (A2) gives, for every measurable set A,

$$\begin{aligned}M_n(x,A) &= \int_A f_{X_n|X_{n-1}}(x_n \mid x;\theta)\,dx_n\\ &\geq \underline{\mathrm{M}}\int_A dx_n\\ &\geq \frac{\underline{\mathrm{M}}}{\bar{M}}\int_A f_{X_n|X_{n-1}}(x_n \mid x^\star;\theta)\,dx_n\\ &= \epsilon M_n(x^\star,A).\end{aligned} \tag{S182}$$

If $\epsilon<1$, Equation (S182) permits the decomposition

$$M_n(x,\cdot) = \epsilon M_n(x^\star,\cdot) + (1-\epsilon)R_n(x,\cdot), \tag{S183}$$

where R_n is a Markov kernel. Therefore, for any x, x',

$$\begin{aligned}M_n(G_n)(x) - M_n(G_n)(x') &= (1-\epsilon)\{R_n(G_n)(x) - R_n(G_n)(x')\},\\ \left|M_n(G_n)(x) - M_n(G_n)(x')\right| &\leq (1-\epsilon)\operatorname{osc}(G_n)\\ &\leq (1-\epsilon)(\bar{G}-\underline{\mathrm{G}}).\end{aligned} \tag{S184}$$

Taking the supremum over x, x' gives

$$\operatorname{osc}\{M_n(G_n)\} \leq (1-\epsilon)(\bar{G}-\underline{\mathrm{G}}). \tag{S185}$$

If $\epsilon = 1$, Equation (S182) implies that $M_n(x, \cdot)$ does not depend on x, so the left side of Equation (S185) is zero and the same bound holds. Taking absolute values in Equation (S180), using Equations (S181)– (S185), and then using Equation (S177), yields

$$
\begin{aligned}
|\Delta_{n,r,i}| &\leq \frac{\operatorname{osc}(H)\operatorname{osc}\{M_n(G_n)\}}{\underline{\mathrm{G}}} \\
&\leq \frac{2(\bar{G} - \underline{\mathrm{G}})}{\underline{\mathrm{G}}} G'(\theta) \begin{cases} (1-\epsilon), & r = 1, \\ (1-\epsilon)^r, & r \geq 2. \end{cases}
\end{aligned} \tag{S186}
$$

When $\epsilon < 1$, take

$$
\varrho = 1 - \epsilon = 1 - \frac{\underline{\mathrm{M}}}{\bar{M}}. \tag{S187}
$$

Then the two cases in Equation (S186) both equal ϱ^r. Summing the squared coordinate bounds and taking square roots give

$$
\begin{aligned}
\|\Delta_{n,r}\|_2 &= \left\{ \sum_{i=1}^{p} |\Delta_{n,r,i}|^2 \right\}^{1/2} \\
&\leq C\sqrt{p}\, G'(\theta)\varrho^r, \qquad 1 \leq r < n.
\end{aligned} \tag{S188}
$$

Step 5: the zero-lag case. Finally, when $r = 0$, Equation (S159) and Equation (S148) give

$$
\begin{aligned}
\|\Delta_{n,0}\|_2 &= \|\rho_n(h_n)\|_2 \\
&\leq \rho_n(\|h_n\|_2) \\
&\leq \sqrt{p}\, G'(\theta) \\
&\leq C\sqrt{p}\, G'(\theta)\varrho^0.
\end{aligned} \tag{S189}
$$

Equations (S188) and (S189) prove Equation (S163). If $\epsilon = 1$, Equation (S180) and the zero oscillation in Equation (S185) show that $\Delta_{n,r} = 0$ for every $r \geq 1$, so any fixed $\varrho \in (0, 1)$ proves the same result. Thus ϱ depends only on the model minorization bounds and is independent of N, J, α, and any lag truncation k. In particular, the factor ϱ^r includes the contraction of the first positive lag; it is not only a tail bound.

□

Lemma S11 (Uniform weak bias of current-particle averages)**.** *Suppose $\theta = \phi$. Under Assumptions (A2), (A3), and (A5), with multinomial resampling, there is $C < \infty$ such that, for every bounded real function f of the current particle location and every $t \geq 1$,*

$$
|\mathbb{E}m(\xi_t)(f) - \rho_t(f)| \leq \frac{C}{J}\operatorname{osc}(f). \tag{S190}
$$

The constant is independent of t and J. Here "current" means measurable with respect to the time-t particle location to which Assumption (A2) applies. In particular, $h_t = h_t(X_t)$ is current at time t. For $s < t$, however, $h_s(\xi_{t,j})$ is selected through the ancestry and is not assumed to be a function of the current location alone. The lemma is therefore not applied directly to a historical h_s at a later time.

Proof. The proof has four stages: reduce the filtered average to the prediction cloud, establish contraction of the normalized semigroup, compute the J^{-1} bias of one particle update, and telescope those updates.

Step 1: prediction-to-filtering reduction. The predictor cloud consists of the current particle locations. A bounded function of the current location is therefore a current-state functional in the Feynman–Kac model already defined in Section S3. This statement does not turn an accumulated weight gradient or an individual ancestral contribution into a function of the current location. The current-location marginal of ρ_t is the filtering distribution, and hence

$$\rho_t(f) = \pi_t(f) = \Psi_t(\eta_t)(f). \tag{S191}$$

Conditionally on the prediction cloud η_t^J, multinomial resampling draws the filtering particles independently from $\Psi_t(\eta_t^J)$. Therefore

$$\begin{aligned}\mathbb{E}\{m(\xi_t)(f) \mid \eta_t^J\} &= \frac{1}{J}\sum_{j=1}^{J} \mathbb{E}\{f(x_{t,j}^F) \mid \eta_t^J\} \\ &= \frac{1}{J}\sum_{j=1}^{J} \Psi_t(\eta_t^J)(f) \\ &= \Psi_t(\eta_t^J)(f).\end{aligned} \tag{S192}$$

Step 2: normalized-semigroup contraction. Take $\Phi_{t,t}$ to be the identity. For $1 \le p \le t$, define

$$\begin{aligned}Q_{t,t}(f)(x) &= G_t(x)f(x), \\ Q_{p,t}(f)(x) &= G_p(x)M_{p+1}\{Q_{p+1,t}(f)\}(x), \qquad p < t.\end{aligned} \tag{S193}$$

The definitions of $\Phi_{p,t}$ and Ψ_t give, for every probability measure μ,

$$\Psi_t\{\Phi_{p,t}(\mu)\}(f) = \frac{\mu Q_{p,t}(f)}{\mu Q_{p,t}(1)}. \tag{S194}$$

For $p < t$,

$$Q_{p,t}(1)(x) = G_p(x)\int f_{X_{p+1}|X_p}(u \mid x;\theta)Q_{p+1,t}(1)(u)\,du. \tag{S195}$$

Thus, for every x, x',

$$\begin{aligned}\frac{Q_{p,t}(1)(x)}{Q_{p,t}(1)(x')} &\le \frac{\bar{G}\bar{M}\int Q_{p+1,t}(1)(u)\,du}{\underline{\mathrm{GM}}\int Q_{p+1,t}(1)(u)\,du} \\ &= \frac{\bar{G}\bar{M}}{\underline{\mathrm{GM}}}.\end{aligned} \tag{S196}$$

For $p = t$, the same ratio is at most $\bar{G}/\underline{\mathrm{G}}$. Put

$$R = \frac{\bar{G}\bar{M}}{\underline{\mathrm{GM}}} \ge 1. \tag{S197}$$

Then

$$\sup_{x,x'} \frac{Q_{p,t}(1)(x)}{Q_{p,t}(1)(x')} \leq R, \qquad 1 \leq p \leq t. \tag{S198}$$

For $p \leq s < t$, the normalized forward kernel is

$$P^{(t)}_{s,s+1}(x, A) = \frac{\displaystyle\int_A f_{X_{s+1}|X_s}(u \mid x; \theta) Q_{s+1,t}(1)(u)\, du}{\displaystyle\int f_{X_{s+1}|X_s}(u \mid x; \theta) Q_{s+1,t}(1)(u)\, du}. \tag{S199}$$

Define, only within this proof,

$$\nu_{s,t}(A) = \frac{\displaystyle\int_A Q_{s+1,t}(1)(u)\, du}{\displaystyle\int Q_{s+1,t}(1)(u)\, du}. \tag{S200}$$

Assumption (A2) gives

$$\begin{aligned} P^{(t)}_{s,s+1}(x, A) &\geq \frac{\underline{\mathrm{M}} \displaystyle\int_A Q_{s+1,t}(1)(u)\, du}{\bar{M} \displaystyle\int Q_{s+1,t}(1)(u)\, du} \\ &= \frac{\underline{\mathrm{M}}}{\bar{M}} \nu_{s,t}(A). \end{aligned} \tag{S201}$$

Consequently, with

$$\varrho = 1 - \frac{\underline{\mathrm{M}}}{2\bar{M}} \in (0, 1), \tag{S202}$$

the Dobrushin coefficient of every kernel in Equation (S199) is at most ϱ. The factor $G_s(x)$ cancels from the normalized conditional expectation, and hence

$$\frac{Q_{p,t}(f)}{Q_{p,t}(1)} = P^{(t)}_{p,p+1} P^{(t)}_{p+1,p+2} \cdots P^{(t)}_{t-1,t} f. \tag{S203}$$

It follows that

$$\operatorname{osc}\left\{\frac{Q_{p,t}(f)}{Q_{p,t}(1)}\right\} \leq \varrho^{t-p} \operatorname{osc}(f). \tag{S204}$$

Step 3: one complete particle update. We next bound one complete selection–mutation update. For $p = 1$, put $\mu = \eta_1 = \pi_0 M_1$. For $p \geq 2$, condition on η^J_{p-1} and put

$$\mu = \Phi_p(\eta^J_{p-1}). \tag{S205}$$

In either case, the particles forming η^J_p are conditionally independent with common law μ. Here $Q_{p,t}$ propagates the local error to time t: q isolates its random normalizer, while d is the centered empirical fluctuation. Within this calculation, put

$$q = \frac{Q_{p,t}(1)}{\mu Q_{p,t}(1)}, \qquad a = \frac{\mu Q_{p,t}(f)}{\mu Q_{p,t}(1)}, \qquad d = \frac{Q_{p,t}(f - a)}{\mu Q_{p,t}(1)}. \tag{S206}$$

Then

$$\mu(q) = 1, \qquad \mu(d) = 0, \tag{S207}$$

and Equation (S194) gives

$$\begin{aligned}
&\Psi_t\{\Phi_{p,t}(\eta_p^J)\}(f) - \Psi_t\{\Phi_{p,t}(\mu)\}(f) \\
&\quad = \frac{\eta_p^J Q_{p,t}(f)}{\eta_p^J Q_{p,t}(1)} - a \\
&\quad = \frac{\eta_p^J Q_{p,t}(f-a)}{\eta_p^J Q_{p,t}(1)} \\
&\quad = \frac{\eta_p^J(d)}{\eta_p^J(q)}.
\end{aligned} \tag{S208}$$

Equation (S198) implies

$$R^{-1} \leq q(x) \leq R, \qquad \eta_p^J(q) \geq R^{-1}, \qquad \|q-1\|_\infty \leq R. \tag{S209}$$

Writing

$$F = \frac{Q_{p,t}(f)}{Q_{p,t}(1)}, \tag{S210}$$

we have $a = \mu(qF)$ and $\mu(q) = 1$. Thus a is an average of F under the probability measure $q\,d\mu$, so

$$\inf F \leq a \leq \sup F. \tag{S211}$$

Since $d = q(F-a)$, Equations (S209), (S211), and (S204) give

$$\begin{aligned}
\|d\|_\infty &\leq R\operatorname{osc}(F) \\
&\leq R\varrho^{t-p}\operatorname{osc}(f).
\end{aligned} \tag{S212}$$

Conditional independence and $\mu(d) = 0$ give

$$\begin{aligned}
\mathbb{E}\{\eta_p^J(d) \mid \mu\} &= 0, \\
\mathbb{E}\{\eta_p^J(d)^2 \mid \mu\} &= \frac{1}{J}\mu(d^2), \\
\mathbb{E}\{(\eta_p^J(q)-1)^2 \mid \mu\} &= \frac{1}{J}\mu\{(q-1)^2\}.
\end{aligned} \tag{S213}$$

Consequently,

$$\begin{aligned}
&\mathbb{E}\left\{ \frac{\eta_p^J(d)}{\eta_p^J(q)} \middle| \mu \right\} \\
&\quad = -\mathbb{E}\left\{ \frac{\eta_p^J(d)\{\eta_p^J(q)-1\}}{\eta_p^J(q)} \middle| \mu \right\}.
\end{aligned} \tag{S214}$$

Conditional Cauchy–Schwarz and Equations (S209)– (S213) yield

$$\left| \mathbb{E}\left\{ \frac{\eta_p^J(d)}{\eta_p^J(q)} \middle| \mu \right\} \right|$$

$$
\begin{aligned}
&\leq R\left[\mathbb{E}\{\eta_p^J(d)^2 \mid \mu\}\mathbb{E}\{(\eta_p^J(q)-1)^2 \mid \mu\}\right]^{1/2} \\
&= \frac{R}{J}\left[\mu(d^2)\mu\{(q-1)^2\}\right]^{1/2} \\
&\leq \frac{C}{J}\varrho^{t-p}\operatorname{osc}(f).
\end{aligned} \tag{S215}
$$

Step 4: telescope over update times. It remains to write the exact full-update telescope. Since $\eta_t = \Phi_{1,t}(\eta_1)$,

$$
\begin{aligned}
&\mathbb{E}\Psi_t(\eta_t^J)(f) - \Psi_t(\eta_t)(f) \\
&\quad = \mathbb{E}\left[\Psi_t\{\Phi_{1,t}(\eta_1^J)\}(f) - \Psi_t\{\Phi_{1,t}(\eta_1)\}(f)\right] \\
&\qquad + \sum_{p=2}^{t}\mathbb{E}\left[\Psi_t\{\Phi_{p,t}(\eta_p^J)\}(f) - \Psi_t\{\Phi_{p,t}(\Phi_p(\eta_{p-1}^J))\}(f)\right].
\end{aligned} \tag{S216}
$$

Indeed, for $2 \leq p \leq t$,

$$
\Phi_{p,t} \circ \Phi_p = \Phi_{p-1,t}, \tag{S217}
$$

so consecutive terms cancel. The only terms left are $\Psi_t(\eta_t^J)(f)$ and $-\Psi_t(\eta_t)(f)$.

Apply Equation (S215) to every summand in Equation (S216). Then

$$
\begin{aligned}
\left|\mathbb{E}\Psi_t(\eta_t^J)(f) - \Psi_t(\eta_t)(f)\right| &\leq \frac{C\operatorname{osc}(f)}{J}\sum_{p=1}^{t}\varrho^{t-p} \\
&\leq \frac{C}{J(1-\varrho)}\operatorname{osc}(f).
\end{aligned} \tag{S218}
$$

Absorb $(1-\varrho)^{-1}$ into C and combine this display with Equations (S191) and (S192). □

Lemma S12 (Full-score summation-by-parts identities)**.** *Suppose $\theta = \phi$ and $\alpha \in [0,1]$. Then the particle and population scores satisfy Equations (S222) and (S223) below. Moreover, for every coordinate i and $t \leq N$,*

$$
\left\|\sum_{k=1}^{t}\alpha^{t-k}h_{k,i}\right\|_\infty \leq G'(\theta)\sum_{k=1}^{t}\alpha^{t-k}. \tag{S219}
$$

Proof. Step 1: particle summation by parts. We first record the summation-by-parts identity used again in the variance proof. Sum Equation (S158) over $n = 1, \ldots, N$ and put $k = n - r$. Since the sums are finite,

$$
\nabla_\theta\hat{\ell}^\alpha(\theta) = \sum_{k=1}^{N}\left[m(\xi_k)(h_k) + \sum_{t=k+1}^{N}\alpha^{t-k}\{m(\xi_t)(h_k) - m(\xi_{t-1})(h_k)\}\right]. \tag{S220}
$$

For each fixed k, collecting the coefficient of every $m(\xi_t)(h_k)$ gives

$$
m(\xi_k)(h_k) + \sum_{t=k+1}^{N}\alpha^{t-k}\{m(\xi_t)(h_k) - m(\xi_{t-1})(h_k)\}
$$

$$= (1-\alpha)\sum_{t=k}^{N-1}\alpha^{t-k}m(\xi_t)(h_k) + \alpha^{N-k}m(\xi_N)(h_k). \tag{S221}$$

Indeed, the coefficients are $1-\alpha$ at $t=k$, $\alpha^{t-k}-\alpha^{t+1-k}=(1-\alpha)\alpha^{t-k}$ for $k<t<N$, and α^{N-k} at $t=N$. Interchanging the finite sums once more yields

$$\nabla_\theta \hat{\ell}^\alpha(\theta) = (1-\alpha)\sum_{t=1}^{N-1} m(\xi_t)\left(\sum_{k=1}^{t}\alpha^{t-k}h_k\right) + m(\xi_N)\left(\sum_{k=1}^{N}\alpha^{N-k}h_k\right). \tag{S222}$$

Step 2: population identity and uniform bound. The same algebra applied to Equation (S162) gives

$$\nabla_\theta \ell^\alpha(\theta) = (1-\alpha)\sum_{t=1}^{N-1} \rho_t\left(\sum_{k=1}^{t}\alpha^{t-k}h_k\right) + \rho_N\left(\sum_{k=1}^{N}\alpha^{N-k}h_k\right). \tag{S223}$$

For every coordinate i, Equation (S148) gives

$$\left\|\sum_{k=1}^{t}\alpha^{t-k}h_{k,i}\right\|_\infty \leq G'(\theta)\sum_{k=1}^{t}\alpha^{t-k}. \tag{S224}$$

Equation (S224) is identical to Equation (S219), completing the proof. □

Claim S1 (Conditional empirical moments)**.** *Let $\mathcal{F}$ be a sigma-field and let μ be an $\mathcal{F}$-measurable probability measure. Suppose that, conditionally on $\mathcal{F}$, $X_1,\ldots,X_J$ are independent with common law μ. The following bounds hold for bounded $\mathcal{F}$-measurable real functions f, f_1, f_2:*

$$\mathbb{E}\left\{\frac{1}{J}\sum_{j=1}^{J}\{f(X_j)-\mu(f)\}\,\middle|\,\mathcal{F}\right\} = 0, \tag{S225}$$

$$\mathbb{E}\left\{\left[\frac{1}{J}\sum_{j=1}^{J}\{f(X_j)-\mu(f)\}\right]^2\,\middle|\,\mathcal{F}\right\} = \frac{\mu\{(f-\mu f)^2\}}{J} \leq \frac{\operatorname{osc}(f)^2}{4J}, \tag{S226}$$

$$\mathbb{E}\left\{\left[\frac{1}{J}\sum_{j=1}^{J}\{f(X_j)-\mu(f)\}\right]^4\,\middle|\,\mathcal{F}\right\} \leq \frac{3\operatorname{osc}(f)^4}{J^2}. \tag{S227}$$

Consequently, for bounded f_1 and f_2,

$$\begin{aligned}&\left\|\left[\frac{1}{J}\sum_{j=1}^{J}\{f_1(X_j)-\mu(f_1)\}\right]\left[\frac{1}{J}\sum_{j=1}^{J}\{f_2(X_j)-\mu(f_2)\}\right]\right\|_{L^2|\mathcal{F}} \\ &\quad\leq \frac{\sqrt{3}}{J}\operatorname{osc}(f_1)\operatorname{osc}(f_2). \end{aligned}\tag{S228}$$

Proof. Put $Y_j = f(X_j) - \mu(f)$. Conditional independence and $\mathbb{E}(Y_j \mid \mathcal{F}) = 0$ give Equations (S225) and (S226). They also give

$$\begin{aligned}
&\mathbb{E}\left\{ \left(\frac{1}{J} \sum_{j=1}^{J} Y_j \right)^4 \middle| \mathcal{F} \right\} \\
&= \frac{1}{J^4} \left\{ J\mu\{(f-\mu f)^4\} + 3J(J-1)\mu\{(f-\mu f)^2\}^2 \right\} \\
&\leq \frac{J + 3J(J-1)}{J^4} \operatorname{osc}(f)^4 \leq \frac{3}{J^2} \operatorname{osc}(f)^4,
\end{aligned} \tag{S229}$$

which is Equation (S227). Conditional Cauchy–Schwarz applied to the two fourth moments proves Equation (S228). ☐

Claim S2 (Normalized-ratio perturbations)**.** *Let μ and ν be probability measures. For $l \in \{0,1\}$, let $a_l > 0$ and b_l be bounded functions such that $\mu(a_l) = 1$, and put*

$$\mathcal{R}_l(\lambda) = \frac{\lambda(b_l)}{\lambda(a_l)}, \qquad D_\mu \mathcal{R}_l = b_l - \mathcal{R}_l(\mu) a_l. \tag{S230}$$

Then

$$\mathcal{R}_l(\nu) - \mathcal{R}_l(\mu) = (\nu - \mu) D_\mu \mathcal{R}_l - \frac{(\nu-\mu)(a_l)}{\nu(a_l)} (\nu - \mu) D_\mu \mathcal{R}_l, \tag{S231}$$

and hence

$$\begin{aligned}
&\{\mathcal{R}_1(\nu) - \mathcal{R}_0(\nu)\} - \{\mathcal{R}_1(\mu) - \mathcal{R}_0(\mu)\} \\
&\quad - (\nu-\mu)\{D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0\} \\
&= -\frac{(\nu-\mu)(a_1 - a_0)}{\nu(a_0)\nu(a_1)} (\nu-\mu) D_\mu \mathcal{R}_1 \\
&\quad - \frac{(\nu-\mu)(a_0)}{\nu(a_0)} (\nu-\mu)\{D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0\}.
\end{aligned} \tag{S232}$$

If $\inf_{l\in\{0,1\}} \nu(a_l) \geq C^{-1}$, then the absolute value of the left side of Equation (S232) is at most

$$\begin{aligned}
&C^2 \left|(\nu-\mu)(a_1 - a_0)\right| \left|(\nu-\mu) D_\mu \mathcal{R}_1\right| \\
&\quad + C \left|(\nu-\mu)(a_0)\right| \left|(\nu-\mu)\{D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0\}\right|.
\end{aligned} \tag{S233}$$

Proof. Since $\mu(a_l) = 1$,

$$\begin{aligned}
\mathcal{R}_l(\nu) - \mathcal{R}_l(\mu) &= \frac{\nu(b_l) - \mathcal{R}_l(\mu)\nu(a_l)}{\nu(a_l)} \\
&= \frac{(\nu-\mu) D_\mu \mathcal{R}_l}{\nu(a_l)} \\
&= (\nu-\mu) D_\mu \mathcal{R}_l - \frac{(\nu-\mu)(a_l)}{\nu(a_l)} (\nu-\mu) D_\mu \mathcal{R}_l,
\end{aligned} \tag{S234}$$

which proves Equation (S231). Moreover,

$$\begin{aligned}
&\frac{(\nu-\mu)(a_1)}{\nu(a_1)} - \frac{(\nu-\mu)(a_0)}{\nu(a_0)} \\
&= \left(1 - \frac{1}{\nu(a_1)}\right) - \left(1 - \frac{1}{\nu(a_0)}\right) \\
&= \frac{\nu(a_1) - \nu(a_0)}{\nu(a_0)\nu(a_1)} \\
&= \frac{(\nu-\mu)(a_1 - a_0)}{\nu(a_0)\nu(a_1)}.
\end{aligned} \tag{S235}$$

Subtracting the cases $l = 0$ and $l = 1$ in Equation (S231) now proves Equation (S232). Taking absolute values there and using $\nu(a_0)^{-1} \le C$ and $\nu(a_1)^{-1} \le C$ proves Equation (S233). This exact identity is the paired specialization of the normalized Feynman–Kac perturbation calculation in Equations (3.1)–(3.4) and Appendix A.3 of Del Moral and Rio (2011). □

Lemma S13 (Particle error and variance of the DMOP-α score)**.** *Suppose $J \ge 2$, $\theta = \phi$, $\alpha \in [0, 1)$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Assume additionally that the bounded measurement-gradient input is the current function $h_n(X_n)$ in Equation (S146). Then*

$$\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\right\|_2^2 \lesssim pG'(\theta)^2 \left\{\frac{N}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2}\right\}, \tag{S236}$$

and

$$\mathrm{Var}\left(\nabla_\theta \hat{\ell}^\alpha(\theta)\right) \lesssim pG'(\theta)^2 \left\{\frac{N}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2}\right\}. \tag{S237}$$

Here the multiplicative constants suppressed by $\lesssim$ depend on the fixed model bounds, but are independent of N, J, and α, and $\mathrm{Var}(Z) = \mathbb{E}\|Z - \mathbb{E}Z\|_2^2$. In particular, if $J \ge N$, the second term is absorbed by the first and

$$\mathrm{Var}\left(\nabla_\theta \hat{\ell}^\alpha(\theta)\right) \lesssim \frac{NpG'(\theta)^2}{J(1-\alpha)^2}. \tag{S238}$$

Proof. The proof keeps three time indices explicit: s is the time at which $h_{s,i}$ is created, t is the local particle-sampling time, and n is the conditional-score time. We first separate terminal resampling, then control predictable contributions for $s \le t$ and $t < s$, and only then sum the local particle errors over t.

Step 1: terminal multinomial resampling. Fix a coordinate i and write

$$Z_i^J = \partial_{\theta_i} \hat{\ell}^\alpha(\theta), \qquad z_i^\alpha = \partial_{\theta_i} \ell^\alpha(\theta). \tag{S239}$$

Equation (S158) and Equation (S148) give

$$\begin{aligned}
|Z_i^J| &\le \sum_{n=1}^N \left\{G'(\theta) + 2G'(\theta) \sum_{r=1}^{n-1} \alpha^r\right\} \\
&\le NG'(\theta)\left\{1 + \frac{2}{1-\alpha}\right\} < \infty.
\end{aligned} \tag{S240}$$

Thus all random variables used below are square integrable.

We first separate the error created by the time-n multinomial resampling. For the prediction paths at time n, write

$$p_{n,a} = \frac{G_n(\xi^P_{n,a})}{\sum_{b=1}^J G_n(\xi^P_{n,b})}, \qquad a = 1, \ldots, J, \tag{S241}$$

and put

$$R^J_{n,i} = \frac{1}{J} \sum_{j=1}^J \left[\sum_{s=1}^n \alpha^{n-s} h_{s,i}(\xi^P_{n,k_j}) - \sum_{a=1}^J p_{n,a} \sum_{s=1}^n \alpha^{n-s} h_{s,i}(\xi^P_{n,a}) \right]. \tag{S242}$$

Conditionally on the prediction paths, the J summands in Equation (S242) are independent and centered. Moreover,

$$\begin{aligned} \operatorname{osc}\left(\sum_{s=1}^n \alpha^{n-s} h_{s,i} \right) &\le 2G'(\theta) \sum_{s=1}^n \alpha^{n-s} \\ &\le \frac{2G'(\theta)}{1-\alpha}. \end{aligned} \tag{S243}$$

It follows from Claim S1 and Equation (S243) that

$$\mathbb{E}\left\{ (R^J_{n,i})^2 \mid \xi^P_{n,1:J} \right\} \le \frac{G'(\theta)^2}{J(1-\alpha)^2}. \tag{S244}$$

Step 2: exact predictable-score decomposition. We next derive the local perturbation bound for the predictable part. Fix $s \le t \le n$ and retain the manuscript's $h_{s,i}$. Define

$$\begin{aligned} q^1_{t,n} &= Q_{t,n}(1), \\ q^0_{t,n} &= \begin{cases} Q_{t,n-1}(1), & t < n, \\ 1, & t = n, \end{cases} \qquad\qquad F_{t,n} = \frac{q^1_{t,n}}{q^0_{t,n}}, \end{aligned} \tag{S245}$$

and, for a probability measure μ on prediction paths through time t,

$$C_{t,n}(\mu) = \frac{\mu(h_{s,i} q^1_{t,n})}{\mu(q^1_{t,n})} - \frac{\mu(h_{s,i} q^0_{t,n})}{\mu(q^0_{t,n})}. \tag{S246}$$

At $t = n$, this is precisely the predictable part of the one-lag difference:

$$\begin{aligned} C_{n,n}\{m(\xi^P_n)\} &= \sum_{a=1}^J p_{n,a} h_{s,i}(\xi^P_{n,a}) - m(\xi^P_n)(h_{s,i}), \\ C_{n,n}(\eta_n) &= \begin{cases} \rho_n(h_{s,i}) - \rho_{n-1}(h_{s,i}), & s < n, \\ \rho_n(h_{n,i}) - \eta_n(h_{n,i}), & s = n. \end{cases} \end{aligned} \tag{S247}$$

When several values of s occur in the same display below, write $C^{(s)}_{t,n}$ for the same functional with the displayed $h_{s,i}$; the superscript records no additional object. The conditional-score decomposition gives the exact identities

$$Z^J_i = \sum_{n=1}^N R^J_{n,i} + \sum_{n=1}^N m(\xi^P_n)(h_{n,i}) + \sum_{n=1}^N \sum_{s=1}^n \alpha^{n-s} C^{(s)}_{n,n}\{m(\xi^P_n)\}, \tag{S248}$$

$$z_i^\alpha = \sum_{n=1}^{N} \eta_n(h_{n,i}) + \sum_{n=1}^{N}\sum_{s=1}^{n} \alpha^{n-s} C_{n,n}^{(s)}(\eta_n). \tag{S249}$$

Indeed, for $s < n$, the sth summand of $R_{n,i}^J$ plus $\alpha^{n-s}C_{n,n}^{(s)}\{m(\xi_n^P)\}$ is

$$\alpha^{n-s}\left\{m(\xi_n)(h_{s,i}) - m(\xi_{n-1})(h_{s,i})\right\}, \tag{S250}$$

while for $s = n$ the three terms are

$$\begin{aligned}\left[m(\xi_n)(h_{n,i}) - \sum_{a=1}^{J} p_{n,a} h_{n,i}(\xi_{n,a}^P)\right] + m(\xi_n^P)(h_{n,i})&\\ + \left[\sum_{a=1}^{J} p_{n,a} h_{n,i}(\xi_{n,a}^P) - m(\xi_n^P)(h_{n,i})\right] &= m(\xi_n)(h_{n,i}).\end{aligned} \tag{S251}$$

This proves Equation (S248); the population identity follows by the same cancellation.

Step 3: continuation when $s \le t$. The recursion for $Q_{t,n}$ and preservation of $h_{s,i}$ after time s give, for $s < t \le n$,

$$C_{t,n}\{\Phi_t(\mu)\} = C_{t-1,n}(\mu). \tag{S252}$$

Indeed, for $l \in \{0, 1\}$,

$$\begin{aligned}\frac{\Phi_t(\mu)(h_{s,i}q_{t,n}^l)}{\Phi_t(\mu)(q_{t,n}^l)} &= \frac{\mu\{G_{t-1}M_t(h_{s,i}q_{t,n}^l)\}}{\mu\{G_{t-1}M_t(q_{t,n}^l)\}}\\ &= \frac{\mu\{h_{s,i}G_{t-1}M_t(q_{t,n}^l)\}}{\mu\{G_{t-1}M_t(q_{t,n}^l)\}}\\ &= \frac{\mu(h_{s,i}q_{t-1,n}^l)}{\mu(q_{t-1,n}^l)}.\end{aligned} \tag{S253}$$

Subtracting the cases $l = 1$ and $l = 0$ proves Equation (S252).

Assumptions (A2) and (A3) imply, exactly as in Equations (S198) and (S204), that

$$\begin{aligned}\sup_{x,x'} \frac{q_{t,n}^0(x)}{q_{t,n}^0(x')} &\le C,\\ \underline{\mathrm{G}} \le F_{t,n}(x) &\le \bar{G},\\ \operatorname{osc}(F_{t,n}) &\le C\varrho^{n-t}.\end{aligned} \tag{S254}$$

Normalize $q_{t,n}^0$ so that $\mu(q_{t,n}^0) = 1$ and write

$$\bar{\mu}(f) = \mu(q_{t,n}^0 f). \tag{S255}$$

Then

$$C_{t,n}(\mu) = \frac{\bar{\mu}(h_{s,i}F_{t,n})}{\bar{\mu}(F_{t,n})} - \bar{\mu}(h_{s,i}). \tag{S256}$$

For constants h_0 and F_0 in the respective ranges of $h_{s,i}$ and $F_{t,n}$,

$$\bar{\mu}(h_{s,i}F_{t,n}) - \bar{\mu}(h_{s,i})\bar{\mu}(F_{t,n})$$

$$= \bar{\mu}\{(h_{s,i} - h_0)(F_{t,n} - F_0)\} - \bar{\mu}(h_{s,i} - h_0)\bar{\mu}(F_{t,n} - F_0). \tag{S257}$$

Equations (S148) and (S254) therefore give

$$|C_{t,n}(\mu)| \leq CG'(\theta)\varrho^{n-t}. \tag{S258}$$

To control the particle perturbation, put

$$\mathcal{R}_l(\lambda) = \frac{\lambda(h_{s,i}q^l_{t,n})}{\lambda(q^l_{t,n})}, \qquad a_l = \frac{q^l_{t,n}}{\mu(q^l_{t,n})}, \qquad l \in \{0,1\}. \tag{S259}$$

Thus $C_{t,n} = \mathcal{R}_1 - \mathcal{R}_0$. With the normalization $\mu(q^0_{t,n}) = 1$, direct differentiation gives

$$\begin{aligned} a_0 &= q^0_{t,n}, & a_1 &= \frac{q^0_{t,n}F_{t,n}}{\bar{\mu}(F_{t,n})}, \\ D_\mu\mathcal{R}_0 &= q^0_{t,n}\{h_{s,i} - \bar{\mu}(h_{s,i})\}, & D_\mu\mathcal{R}_1 &= a_1\left\{h_{s,i} - \frac{\bar{\mu}(h_{s,i}F_{t,n})}{\bar{\mu}(F_{t,n})}\right\}. \end{aligned} \tag{S260}$$

Subtracting the two expressions in Equation (S260) gives the exact first variation

$$\begin{aligned} D_\mu C_{t,n} &= D_\mu\mathcal{R}_1 - D_\mu\mathcal{R}_0 \\ &= (a_1 - a_0)\{h_{s,i} - \mathcal{R}_1(\mu)\} + a_0\{\mathcal{R}_0(\mu) - \mathcal{R}_1(\mu)\}. \end{aligned} \tag{S261}$$

Equations (S254), (S148), and (S258) therefore imply

$$\begin{aligned} \operatorname{osc}(a_0) &\leq C, \\ \operatorname{osc}(a_1 - a_0) &\leq C\varrho^{n-t}, \\ \operatorname{osc}(D_\mu\mathcal{R}_1) &\leq CG'(\theta), \\ \operatorname{osc}(D_\mu C_{t,n}) &\leq CG'(\theta)\varrho^{n-t}. \end{aligned} \tag{S262}$$

The uniform ratio bounds also give $\inf_l \nu(a_l) \geq C^{-1}$. Apply Claim S2 with $b_l = a_l h_{s,i}$. Equations (S233) and (S262) give

$$\begin{aligned} &|C_{t,n}(\nu) - C_{t,n}(\mu) - (\nu - \mu)D_\mu C_{t,n}| \\ &\quad \leq C^2\,|(\nu - \mu)(a_1 - a_0)|\,|(\nu - \mu)D_\mu\mathcal{R}_1| \\ &\qquad + C\,|(\nu - \mu)(a_0)|\,|(\nu - \mu)D_\mu C_{t,n}|\,. \end{aligned} \tag{S263}$$

In each product on the right, one oscillation is bounded by $C\varrho^{n-t}$ and the other by $CG'(\theta)$. Thus the factor ϱ^{n-t} follows from the displayed two-ratio cancellation, rather than from a mixing assertion for the historical function $h_{s,i}$.

Step 4: telescope over particle updates. We now write the full local telescope before summing any norms. Put

$$\eta^J_t = m(\xi^P_t), \qquad \mu^J_t = \begin{cases} \eta_1, & t = 1, \\ \Phi_t(\eta^J_{t-1}), & t \geq 2. \end{cases} \tag{S264}$$

In this display, Φ_t denotes the canonical path-space lift of the same predictor update: the potential is evaluated at the last location, multinomial selection chooses a complete path, and M_t appends the new location. Its last-coordinate marginal is the predictor update already defined in Section S3. Thus, conditionally on the preceding prediction cloud, the J particles forming η_t^J are independent with common law μ_t^J. For the current predictor term in Equation (S248), define

$$H_{t,n}(\mu) = \Phi_{t,n}(\mu)(h_{n,i}), \qquad 1 \le t \le n, \tag{S265}$$

where $\Phi_{n,n}$ is the identity. For $t < s$, extend the already defined continuation functional by

$$C_{t,n}^{(s)}(\mu) = C_{s,n}^{(s)}\{\Phi_{t,s}(\mu)\}. \tag{S266}$$

Equations (S252) and (S217) then give the exact telescopes

$$\begin{aligned} &H_{n,n}(\eta_n^J) - H_{n,n}(\eta_n) \\ &\quad = \sum_{t=1}^{n} \left[H_{t,n}(\eta_t^J) - H_{t,n}(\mu_t^J)\right], \end{aligned} \tag{S267}$$

$$\begin{aligned} &C_{n,n}^{(s)}(\eta_n^J) - C_{n,n}^{(s)}(\eta_n) \\ &\quad = \sum_{t=1}^{n} \left[C_{t,n}^{(s)}(\eta_t^J) - C_{t,n}^{(s)}(\mu_t^J)\right]. \end{aligned} \tag{S268}$$

For example, the second term in the tth summand of Equation (S268) is $C_{t-1,n}^{(s)}(\eta_{t-1}^J)$ for $t \ge 2$, so consecutive terms cancel. At $t = 1$, $\mu_1^J = \eta_1$, and the remaining endpoints are exactly those displayed on the left. The same verification proves Equation (S267).

Step 5: continuation when $t < s$. We now propagate a current function from its creation time. For $t \le n$, define

$$\widetilde{Q}_{t,n}(f) = \begin{cases} f, & t = n, \\ Q_{t,n-1}\{M_n(f)\}, & t < n. \end{cases} \tag{S269}$$

The definition of $\Phi_{t,n}$ gives

$$\Phi_{t,n}(\mu)(f) = \frac{\mu\widetilde{Q}_{t,n}(f)}{\mu\widetilde{Q}_{t,n}(1)}. \tag{S270}$$

For $t < n$, $\widetilde{Q}_{t,n}(1) = Q_{t,n-1}(1)$, so Equation (S198) gives the first bound below. For every $t \le r < n$, the normalized one-step kernel below satisfies

$$\begin{aligned} &\frac{M_{r+1}(x, du)\widetilde{Q}_{r+1,n}(1)(u)}{M_{r+1}\{\widetilde{Q}_{r+1,n}(1)\}(x)} \\ &\quad \ge \frac{\underline{M}}{\bar{M}} \frac{\widetilde{Q}_{r+1,n}(1)(u)\, du}{\int \widetilde{Q}_{r+1,n}(1)(v)\, dv}. \end{aligned} \tag{S271}$$

Its Dobrushin coefficient is therefore at most ϱ. Composing these kernels exactly as in Equations (S203) and (S204) gives

$$\sup_{x,x'} \frac{\widetilde{Q}_{t,n}(1)(x)}{\widetilde{Q}_{t,n}(1)(x')} \le C,$$

$$\operatorname{osc}\left\{\frac{\widetilde{Q}_{t,n}(f)}{\widetilde{Q}_{t,n}(1)}\right\} \le \varrho^{n-t}\operatorname{osc}(f). \tag{S272}$$

The same display is immediate at $t = n$.

For the current predictor continuation in Equation (S265), differentiating its normalized ratio gives

$$D_\mu H_{t,n}(x) = \frac{\widetilde{Q}_{t,n}(1)(x)}{\mu\widetilde{Q}_{t,n}(1)}\left\{\frac{\widetilde{Q}_{t,n}(h_{n,i})(x)}{\widetilde{Q}_{t,n}(1)(x)} - H_{t,n}(\mu)\right\}. \tag{S273}$$

Equations (S272) and (S148) imply

$$\operatorname{osc}\{D_\mu H_{t,n}\} \le CG'(\theta)\varrho^{n-t}. \tag{S274}$$

Applying the single-ratio identity (S231) to $H_{t,n}$ shows that

$$\begin{aligned} &|H_{t,n}(\nu) - H_{t,n}(\mu) - (\nu-\mu)D_\mu H_{t,n}| \\ &\quad \le C\,|(\nu-\mu)(a_H)|\,|(\nu-\mu)D_\mu H_{t,n}|, \end{aligned} \tag{S275}$$

where the normalized denominator $a_H = \widetilde{Q}_{t,n}(1)/\{\mu\widetilde{Q}_{t,n}(1)\}$ satisfies $\mu(a_H) = 1$ and $C^{-1} \le a_H \le C$, so $\nu(a_H) \ge C^{-1}$. The two empirical factors in Equation (S231) have oscillations at most C and $CG'(\theta)\varrho^{n-t}$, respectively, by Equations (S272) and (S274).

For $t < s \le n$, put $\mu_s = \Phi_{t,s}(\mu)$ and

$$d_{s,n} = D_{\mu_s} C^{(s)}_{s,n}. \tag{S276}$$

At time s, the functions $h_{s,i}$, $q^0_{s,n}$, and $q^1_{s,n}$ depend on the current location. Hence Equation (S261) shows that $d_{s,n}$ is a current function satisfying

$$\operatorname{osc}(d_{s,n}) \le CG'(\theta)\varrho^{n-s}. \tag{S277}$$

Differentiating Equation (S266) and using Equation (S270) give

$$D_\mu C^{(s)}_{t,n}(x) = \frac{\widetilde{Q}_{t,s}(1)(x)}{\mu\widetilde{Q}_{t,s}(1)}\left\{\frac{\widetilde{Q}_{t,s}(d_{s,n})(x)}{\widetilde{Q}_{t,s}(1)(x)} - \mu_s(d_{s,n})\right\}. \tag{S278}$$

Equation (S272) applied from t to s, followed by Equation (S277), gives

$$\begin{aligned} \operatorname{osc}\left\{\frac{\widetilde{Q}_{t,s}(d_{s,n})}{\widetilde{Q}_{t,s}(1)}\right\} &\le \varrho^{s-t}\operatorname{osc}(d_{s,n}) \\ &\le CG'(\theta)\varrho^{s-t}\varrho^{n-s} \\ &= CG'(\theta)\varrho^{n-t}. \end{aligned} \tag{S279}$$

The normalizer bound in Equation (S272) now proves

$$\begin{aligned} &\operatorname{osc}\{D_\mu H_{t,n}\} \le CG'(\theta)\varrho^{n-t}, \\ &\operatorname{osc}\{D_\mu C^{(s)}_{t,n}\} \le CG'(\theta)\varrho^{n-t}, \qquad t < s \le n. \end{aligned} \tag{S280}$$

It remains to verify the quadratic remainder for $C_{t,n}^{(s)}$ when $t < s$. Its two normalized ratios are

$$\mathcal{R}_{l,t,s}(\lambda) = \frac{\lambda\widetilde{Q}_{t,s}(h_{s,i}q_{s,n}^l)}{\lambda\widetilde{Q}_{t,s}(q_{s,n}^l)}, \qquad l \in \{0,1\}, \tag{S281}$$

and $C_{t,n}^{(s)} = \mathcal{R}_{1,t,s} - \mathcal{R}_{0,t,s}$. The semigroup recursion gives

$$\widetilde{Q}_{t,s}(q_{s,n}^l) = q_{t,n}^l, \qquad l \in \{0,1\}. \tag{S282}$$

Consequently, the normalized denominator factors are precisely the a_l in Equation (S259), and

$$\operatorname{osc}(a_0) \le C, \qquad \operatorname{osc}(a_1 - a_0) \le C\varrho^{n-t}, \qquad \inf_l \nu(a_l) \ge C^{-1}. \tag{S283}$$

The first variation of the difference is $D_\mu C_{t,n}^{(s)}$, already bounded in Equation (S280). For the first single ratio, the chain rule gives

$$D_\mu \mathcal{R}_{1,t,s}(x) = \frac{\widetilde{Q}_{t,s}(1)(x)}{\mu\widetilde{Q}_{t,s}(1)} \left[\frac{\widetilde{Q}_{t,s}(D_{\mu_s}\mathcal{R}_{1,s,s})(x)}{\widetilde{Q}_{t,s}(1)(x)} - \mu_s(D_{\mu_s}\mathcal{R}_{1,s,s})\right]. \tag{S284}$$

Consequently,

$$\begin{aligned} \operatorname{osc}(D_\mu \mathcal{R}_{1,t,s}) &\le C\varrho^{s-t} \operatorname{osc}(D_{\mu_s}\mathcal{R}_{1,s,s}) \\ &\le CG'(\theta). \end{aligned} \tag{S285}$$

The last inequality follows directly from Equation (S230), the bounded normalizer ratios, and $\operatorname{osc}(h_{s,i}) \le 2G'(\theta)$. Apply Claim S2 with

$$a_l = \frac{\widetilde{Q}_{t,s}(q_{s,n}^l)}{\mu\widetilde{Q}_{t,s}(q_{s,n}^l)}, \qquad b_l = \frac{\widetilde{Q}_{t,s}(h_{s,i}q_{s,n}^l)}{\mu\widetilde{Q}_{t,s}(q_{s,n}^l)}. \tag{S286}$$

Equations (S233), (S283), (S280), and (S285) give

$$\begin{aligned} &\left|C_{t,n}^{(s)}(\nu) - C_{t,n}^{(s)}(\mu) - (\nu-\mu)D_\mu C_{t,n}^{(s)}\right| \\ &\quad \le C^2 \left|(\nu-\mu)(a_1 - a_0)\right| \left|(\nu-\mu)D_\mu \mathcal{R}_{1,t,s}\right| \\ &\qquad + C\left|(\nu-\mu)(a_0)\right| \left|(\nu-\mu)D_\mu C_{t,n}^{(s)}\right|. \end{aligned} \tag{S287}$$

In each product on the right, one oscillation is bounded by $C\varrho^{n-t}$ and the other by $CG'(\theta)$. Together with Equations (S263) and (S275), this proves the required first- and second-order continuation bounds for every t, s, n.

Step 6: sum deterministic coefficients before norms. Equation (S146) shows why no additional mixing coefficient is required for the accumulated weight gradient: the previously accumulated component is multiplied by α, while every new input is a bounded function $h_s(X_s)$ of the mixing location. The proof now sums those explicit coefficients before taking norms. After collecting all score contributions at their local sampling time t, those with $s \le t \le n$ have total coefficient

$$CG'(\theta) \sum_{s=1}^{t} \sum_{n=t}^{N} \alpha^{n-s} \varrho^{n-t}$$

$$
\begin{aligned}
&= CG'(\theta)\left(\sum_{a=0}^{t-1}\alpha^a\right)\left(\sum_{b=0}^{N-t}(\alpha\varrho)^b\right)\\
&\le \frac{CG'(\theta)}{(1-\alpha)(1-\alpha\varrho)}.
\end{aligned}
\tag{S288}
$$

Those with $t < s \le n$ have total coefficient

$$
\begin{aligned}
&CG'(\theta)\sum_{s=t+1}^{N}\sum_{n=s}^{N}\alpha^{n-s}\varrho^{s-t}\varrho^{n-s}\\
&\quad\le CG'(\theta)\left(\sum_{a=1}^{\infty}\varrho^a\right)\left(\sum_{b=0}^{\infty}(\alpha\varrho)^b\right)\\
&\quad= \frac{CG'(\theta)\varrho}{(1-\varrho)(1-\alpha\varrho)}.
\end{aligned}
\tag{S289}
$$

The current predictor terms in Equation (S267) contribute

$$
CG'(\theta)\sum_{n=t}^{N}\varrho^{n-t} \le \frac{CG'(\theta)}{1-\varrho}. \tag{S290}
$$

Since $1-\alpha\varrho \ge 1-\varrho$, the oscillation of the aggregated first-order local function, and the coefficient of its quadratic remainder, are at most

$$
\frac{CG'(\theta)}{1-\alpha}. \tag{S291}
$$

Step 7: local particle errors. We next define the two local terms rather than citing a fixed-terminal formula for their sum. For the signed empirical error $\eta_t^J - \mu_t^J$, put

$$
\Gamma_{t,i}^J = (\eta_t^J - \mu_t^J)\left[\sum_{n=t}^{N} D_{\mu_t^J}H_{t,n} + \sum_{n=t}^{N}\sum_{s=1}^{n}\alpha^{n-s}D_{\mu_t^J}C_{t,n}^{(s)}\right], \tag{S292}
$$

$$
\begin{aligned}
B_{t,i}^J &= \sum_{n=t}^{N}\Big[H_{t,n}(\eta_t^J) - H_{t,n}(\mu_t^J) - (\eta_t^J - \mu_t^J)D_{\mu_t^J}H_{t,n}\Big]\\
&\quad+ \sum_{n=t}^{N}\sum_{s=1}^{n}\alpha^{n-s}\Big[C_{t,n}^{(s)}(\eta_t^J) - C_{t,n}^{(s)}(\mu_t^J) - (\eta_t^J - \mu_t^J)D_{\mu_t^J}C_{t,n}^{(s)}\Big].
\end{aligned}
\tag{S293}
$$

Substitute the telescopes Equations (S267) and (S268) into the difference of Equations (S248) and (S249), and then use Equations (S292) and (S293). Term-by-term collection gives the exact identity

$$
Z_i^J - z_i^\alpha = \sum_{n=1}^{N} R_{n,i}^J + \sum_{t=1}^{N}\Gamma_{t,i}^J + \sum_{t=1}^{N} B_{t,i}^J, \tag{S294}
$$

which is the local perturbation construction in Section 5, in particular Equation (5.5), and Appendix A.1 of Del Moral and Rio (2011), with the DMOP sums taken only after the local fields have been formed. Conditionally on the information just before the combined time-t multinomial-selection and mutation block, η_t^J is the empirical measure of J independent draws with common law μ_t^J. Claim S1, the continuation bounds in Steps 3 and 5, and Equation (S291) therefore give

$$\begin{aligned}
&\mathbb{E}(\Gamma_{t,i}^J \mid \mathcal{F}_{t-1}) = 0, \\
&\mathbb{E}\{(\Gamma_{t,i}^J)^2 \mid \mathcal{F}_{t-1}\} \leq \frac{1}{4J} \operatorname{osc} \left[\sum_{n=t}^{N} D_{\mu_t^J} H_{t,n} + \sum_{n=t}^{N} \sum_{s=1}^{n} \alpha^{n-s} D_{\mu_t^J} C_{t,n}^{(s)} \right]^2 \\
&\qquad\qquad\qquad\qquad \leq \frac{CG'(\theta)^2}{J(1-\alpha)^2}.
\end{aligned} \tag{S295}$$

Here $\mathcal{F}_{t-1}$ is the sigma-field just before that combined selection–mutation block: η_{t-1}^J is measurable, but the parent selections and mutations used to construct η_t^J have not yet been revealed. Equation (S228) applied to the products in Equations (S263), (S275), and (S287), followed by the three coefficient sums in Step 6, gives

$$\begin{aligned}
\|B_{t,i}^J\|_{L^2|\mathcal{F}_{t-1}} &\leq \frac{C}{J} \left\{ \frac{G'(\theta)}{1-\varrho} + \frac{G'(\theta)}{(1-\alpha)(1-\alpha\varrho)} + \frac{G'(\theta)\varrho}{(1-\varrho)(1-\alpha\varrho)} \right\} \\
&\leq \frac{CG'(\theta)}{J(1-\alpha)}.
\end{aligned} \tag{S296}$$

The same bound holds for the unconditional L^2 norm.

Step 8: martingale and remainder summation. Expose the terminal multinomial selections in chronological order. Before the time-n selection, the prediction cloud $\xi_{n,1:J}^P$ and every $R_{m,i}^J$ with $m < n$ are measurable, while Equation (S242) gives conditional mean zero for $R_{n,i}^J$. Hence

$$\mathbb{E}(R_{m,i}^J R_{n,i}^J) = 0, \qquad m < n. \tag{S297}$$

Likewise, expose the prediction draws in chronological order. The sigma-field $\mathcal{F}_{t-1}$ in Step 7 contains every $\Gamma_{r,i}^J$ with $r < t$, and Equation (S295) gives

$$\mathbb{E}(\Gamma_{r,i}^J \Gamma_{t,i}^J) = 0, \qquad r < t. \tag{S298}$$

The two families need not be mutually orthogonal because the selection at time n also enters the next prediction cloud. Applying $\|A + B\|_{L^2}^2 \leq 2\|A\|_{L^2}^2 + 2\|B\|_{L^2}^2$ first, and then martingale orthogonality separately within each family, gives

$$\begin{aligned}
\left\| \sum_{n=1}^{N} R_{n,i}^J + \sum_{t=1}^{N} \Gamma_{t,i}^J \right\|_{L^2}^2 &\leq 2 \sum_{n=1}^{N} \|R_{n,i}^J\|_{L^2}^2 + 2 \sum_{t=1}^{N} \|\Gamma_{t,i}^J\|_{L^2}^2 \\
&\leq \frac{CNG'(\theta)^2}{J(1-\alpha)^2}.
\end{aligned} \tag{S299}$$

For the quadratic terms, Minkowski's inequality gives

$$\left\| \sum_{t=1}^{N} B_{t,i}^J \right\|_{L^2} \leq \sum_{t=1}^{N} \|B_{t,i}^J\|_{L^2}$$

$$\leq \frac{CNG'(\theta)}{J(1-\alpha)}. \tag{S300}$$

Combining Equations (S294), (S299), and (S300), and using $(a+b)^2 \leq 2a^2 + 2b^2$, yields

$$\mathbb{E}|Z_i^J - z_i^\alpha|^2 \leq CG'(\theta)^2 \left\{ \frac{N}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2} \right\}. \tag{S301}$$

Summing over $i = 1, \ldots, p$ proves Equation (S236). Since z_i^α is deterministic,

$$\mathrm{Var}(Z_i^J) = \inf_{a \in \mathbb{R}} \mathbb{E}|Z_i^J - a|^2 \leq \mathbb{E}|Z_i^J - z_i^\alpha|^2. \tag{S302}$$

Summing this inequality over the coordinates proves Equation (S237). If $J \geq N$, then $N^2/J^2 \leq N/J$, which proves Equation (S238). □

Corollary S1 (Particle bias of the DMOP-α score)**.** *Under the conditions of Lemma S13, for* $\alpha \in [0, 1)$,

$$\left\| \mathbb{E}\nabla_\theta \hat{\ell}_n^\alpha(\theta) - \nabla_\theta \ell_n^\alpha(\theta) \right\|_2 \leq \frac{C\sqrt{p}\, G'(\theta)}{J(1-\alpha)}, \tag{S303}$$

$$\left\| \mathbb{E}\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta) \right\|_2 \leq \frac{CN\sqrt{p}\, G'(\theta)}{J(1-\alpha)}. \tag{S304}$$

Proof. Fix a coordinate i and a conditional-score time n. Restrict the exact local decomposition in Equation (S294) to this value of n. The terminal resampling term and every first-order local field have conditional mean zero. The quadratic ratio remainders are the only terms with nonzero expectation. Equations (S263), (S280), and (S228) therefore give

$$\begin{aligned} &\left| \mathbb{E}\partial_{\theta_i} \hat{\ell}_n^\alpha(\theta) - \partial_{\theta_i} \ell_n^\alpha(\theta) \right| \\ &\quad \leq \frac{CG'(\theta)}{J} \left\{ \sum_{t=1}^{n} \varrho^{n-t} + \sum_{t=1}^{n} \sum_{s=1}^{t} \alpha^{n-s} \varrho^{n-t} + \sum_{t=1}^{n-1} \sum_{s=t+1}^{n} \alpha^{n-s} \varrho^{s-t} \varrho^{n-s} \right\}. \end{aligned} \tag{S305}$$

The three sums in Equation (S305) satisfy

$$\begin{aligned} \sum_{t=1}^{n} \varrho^{n-t} &\leq \frac{1}{1-\varrho}, \\ \sum_{t=1}^{n} \sum_{s=1}^{t} \alpha^{n-s} \varrho^{n-t} &\leq \left\{ \sum_{a=0}^{\infty} (\alpha\varrho)^a \right\} \left\{ \sum_{b=0}^{\infty} \alpha^b \right\} = \frac{1}{(1-\alpha\varrho)(1-\alpha)}, \\ \sum_{t=1}^{n-1} \sum_{s=t+1}^{n} \alpha^{n-s} \varrho^{s-t} \varrho^{n-s} &\leq \left\{ \sum_{a=1}^{\infty} \varrho^a \right\} \left\{ \sum_{b=0}^{\infty} (\alpha\varrho)^b \right\} = \frac{\varrho}{(1-\varrho)(1-\alpha\varrho)}. \end{aligned} \tag{S306}$$

Since $1 - \alpha\varrho \geq 1 - \varrho$, substitution into Equation (S305) gives

$$\left| \mathbb{E}\partial_{\theta_i} \hat{\ell}_n^\alpha(\theta) - \partial_{\theta_i} \ell_n^\alpha(\theta) \right| \leq \frac{CG'(\theta)}{J(1-\alpha)}. \tag{S307}$$

Taking the Euclidean norm over the coordinates proves Equation (S303). Finally, the triangle inequality gives

$$\begin{aligned}
&\left\|\mathbb{E}\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\right\|_2 \\
&\quad \le \sum_{n=1}^{N} \left\|\mathbb{E}\nabla_\theta \hat{\ell}_n^\alpha(\theta) - \nabla_\theta \ell_n^\alpha(\theta)\right\|_2 \le \frac{CN\sqrt{p}\,G'(\theta)}{J(1-\alpha)},
\end{aligned} \tag{S308}$$

which proves Equation (S304). □

Lemma S14 (Population discounting bias)**.** *Under the conditions of Lemma S10, suppose $\theta = \phi$ and $\alpha \in [0,1]$. Then*

$$\begin{aligned}
\|\nabla_\theta \ell_n^\alpha(\theta) - \nabla_\theta \ell_n(\theta)\|_2 &\le C\sqrt{p}\,G'(\theta) \sum_{r=1}^{n-1} (1-\alpha^r)\varrho^r \\
&\le C\sqrt{p}\,G'(\theta) \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}.
\end{aligned} \tag{S309}$$

Proof. Step 1: exact lagwise difference. Equation (S162) at α and at $\alpha = 1$ gives

$$\begin{aligned}
\nabla_\theta \ell_n(\theta) - \nabla_\theta \ell_n^\alpha(\theta) &= \sum_{r=0}^{n-1} (1-\alpha^r)\Delta_{n,r} \\
&= \sum_{r=1}^{n-1} (1-\alpha^r)\Delta_{n,r},
\end{aligned} \tag{S310}$$

because $1 - \alpha^0 = 0$.

Step 2: apply forgetting and sum. Taking norms, applying the triangle inequality and Lemma S10, and then extending the nonnegative scalar sum to infinity give

$$\begin{aligned}
\|\nabla_\theta \ell_n(\theta) - \nabla_\theta \ell_n^\alpha(\theta)\|_2 &\le \sum_{r=1}^{n-1} (1-\alpha^r)\|\Delta_{n,r}\|_2 \\
&\le C\sqrt{p}\,G'(\theta) \sum_{r=1}^{n-1} (1-\alpha^r)\varrho^r \\
&\le C\sqrt{p}\,G'(\theta) \sum_{r=1}^{\infty} (1-\alpha^r)\varrho^r \\
&= C\sqrt{p}\,G'(\theta) \left\{ \sum_{r=1}^{\infty} \varrho^r - \sum_{r=1}^{\infty} (\alpha\varrho)^r \right\} \\
&= C\sqrt{p}\,G'(\theta) \left\{ \frac{\varrho}{1-\varrho} - \frac{\alpha\varrho}{1-\alpha\varrho} \right\} \\
&= C\sqrt{p}\,G'(\theta) \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}.
\end{aligned} \tag{S311}$$

□

Theorem S5 (MSE decomposition and bound for DMOP-α)**.** *Suppose the observations are fixed, $J \geq 2$, $\theta = \phi$, $\alpha \in [0, 1)$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. The bounded measurement-gradient input in Assumption (A3) is the current function $h_n(X_n)$ in Equation (S146). Then*

$$\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \lesssim pG'(\theta)^2 \left[\frac{N}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2} + \left\{\sum_{r=1}^{N-1}(N-r)(1-\alpha^r)\varrho^r\right\}^2\right]. \tag{S312}$$

Moreover, the exact bias–variance decomposition is

$$\begin{aligned}\mathbb{E}&\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \\ &= \mathrm{Var}\left\{\nabla_\theta \hat{\ell}^\alpha(\theta)\right\} + \left\|\mathbb{E}\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta) + \nabla_\theta \ell^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2,\end{aligned} \tag{S313}$$

where the population discounting term satisfies

$$\begin{aligned}\|\nabla_\theta \ell^\alpha(\theta) - \nabla_\theta \ell(\theta)\|_2 &\leq C\sqrt{p}\, G'(\theta) \sum_{r=1}^{N-1}(N-r)(1-\alpha^r)\varrho^r \\ &\leq CN\sqrt{p}\, G'(\theta)\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}.\end{aligned} \tag{S314}$$

The factor $\varrho \in (0, 1)$ is the one in Lemma S10 and is independent of N, J, and α.

Proof. Step 1: exact bias–variance identity. For a square-integrable random vector Z and a deterministic vector z, direct expansion gives

$$\mathbb{E}\|Z - z\|_2^2 = \mathbb{E}\|Z - \mathbb{E}Z\|_2^2 + \|\mathbb{E}Z - z\|_2^2, \tag{S315}$$

because

$$\mathbb{E}\left[(Z - \mathbb{E}Z)^\mathsf{T}(\mathbb{E}Z - z)\right] = \mathbb{E}(Z - \mathbb{E}Z)^\mathsf{T}(\mathbb{E}Z - z) = 0. \tag{S316}$$

Apply Equation (S315) with $Z = \nabla_\theta \hat{\ell}^\alpha(\theta)$ and $z = \nabla_\theta \ell(\theta)$, and add and subtract $\nabla_\theta \ell^\alpha(\theta)$ inside the deterministic norm. This gives Equation (S313).

Step 2: particle error. Lemma S13 gives the direct particle error

$$\begin{aligned}\mathbb{E}&\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\right\|_2^2 \\ &\lesssim pG'(\theta)^2\left\{\frac{N}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2}\right\}.\end{aligned} \tag{S317}$$

Step 3: population discounting error. Summing the first bound in Lemma S14 over $n = 1, \ldots, N$, using the triangle inequality, and interchanging the two finite sums give

$$\|\nabla_\theta \ell^\alpha(\theta) - \nabla_\theta \ell(\theta)\|_2 \leq C\sqrt{p}\, G'(\theta) \sum_{n=1}^{N}\sum_{r=1}^{n-1}(1-\alpha^r)\varrho^r$$

$$
\begin{aligned}
&= C\sqrt{p}\,G'(\theta)\sum_{r=1}^{N-1}(N-r)(1-\alpha^r)\varrho^r \\
&\le CN\sqrt{p}\,G'(\theta)\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)},
\end{aligned}
\tag{S318}
$$

which is Equation (S314).

Step 4: combine the two errors. Pointwise, $(a+b)^2 \le 2a^2 + 2b^2$ and the triangle inequality give

$$
\begin{aligned}
&\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \\
&\quad \le 2\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\right\|_2^2 + 2\left\|\nabla_\theta \ell^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2.
\end{aligned}
\tag{S319}
$$

Take expectations in Equation (S319) and substitute Equation (S317) and Equation (S314). This proves Equation (S312). The exact bias–variance identity was proved independently at the beginning of the argument; it does not create an additional power of N. □

Remark 1 (Neutral-model check)**.** *The dependence on α in Lemma S13 is conservative but cannot be replaced by a bound involving only the current particle cloud. For example, take iid* Uniform$[0,1]$ *states, constant weights at $\theta = \phi = 0$, and*

$$
\Pr_\theta(Y_n = 1 \mid X_n = x) = \frac{1}{2} + \frac{\theta}{4}(2x-1), \qquad y_n^* = 1. \tag{S320}
$$

Then $h_n = (2X_n - 1)/2$ has variance $\sigma^2 = 1/12$, and direct conditional multinomial calculation gives

$$
\operatorname{Var}\left\{\nabla_\theta \hat{\ell}^\alpha(\theta)\right\} = \frac{\sigma^2}{J}\left[N(2 - J^{-1}) + \sum_{r=1}^{N-1}(N-r)\alpha^{2r}(1-J^{-1})^{r+1}\right]. \tag{S321}
$$

*Thus the variance has order N/J for every fixed $\alpha < 1$, and its genealogical contribution has order $N/\{J(1-\alpha^2)\}$ before horizon saturation. At $\alpha = 0$, this agrees with the DMOP-*0 *bound preceding Theorem S4. The example is consistent with Lemma S13, but shows why the sharper $N/\{J(1-\alpha^2)\}$ rate would require additional joint-lineage control.*

Remark 2 (Choice of α in the fixed-observation upper bound)**.** *The identity $1 - \alpha^r = (1-\alpha)\sum_{q=0}^{r-1}\alpha^q$ and $1 - \alpha^q \le q(1-\alpha)$ imply, uniformly in N,*

$$
\begin{aligned}
0 &\le (1-\alpha)\sum_{r=1}^{N-1}(N-r)r\varrho^r - \sum_{r=1}^{N-1}(N-r)(1-\alpha^r)\varrho^r \\
&\le N(1-\alpha)^2\frac{\varrho^2}{(1-\varrho)^3}, \\
\sum_{r=1}^{N-1}&(N-r)(1-\alpha^r)\varrho^r \le N(1-\alpha)\frac{\varrho}{(1-\varrho)^2}.
\end{aligned}
\tag{S322}
$$

No horizon-saturation argument is used for this population term. Applying the last inequality to Theorem S5 gives the following coarsening of its fixed-observation upper bound, with fixed model factors suppressed:

$$
\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2
$$

$$\lesssim pG'(\theta)^2 \left\{ \frac{1}{(1-\alpha)^2} \left(\frac{N}{J} + \frac{N^2}{J^2} \right) + N^2(1-\alpha)^2 \right\}. \tag{S323}$$

Differentiation gives the interior rate-balancing choice and its value:

$$1-\alpha \asymp \left(\frac{1}{NJ} + \frac{1}{J^2} \right)^{1/4}, \tag{S324}$$

$$\mathbb{E} \left\| \nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta) \right\|_2^2 \lesssim pG'(\theta)^2 N \left(\frac{N}{J} + \frac{N^2}{J^2} \right)^{1/2}. \tag{S325}$$

The factor $(1-\alpha)^{-1}$ in Equation (S323) enters through the replacement of finite sums $\sum_{q=0}^{m} \alpha^q$ in the proof of Lemma S13. Hence the effective-memory interpretation of Equation (S324) requires $(1-\alpha)^{-1} \lesssim N$. At the displayed choice this condition is equivalent, up to fixed factors, to $N^3/J + N^4/J^2 \gtrsim 1$. It holds throughout $J \le N$, where Equation (S325) is of order $pG'(\theta)^2 N^2/J$, and for $N \le J \lesssim N^3$, where it is of order $pG'(\theta)^2 N^{3/2}/\sqrt{J}$. When $J \gg N^3$, the proposed choice has $(1-\alpha)^{-1} \gg N$; the finite sums in the proof of Lemma S13 must then be retained rather than replaced by $(1-\alpha)^{-1}$, and Equation (S325) is not asserted as the optimized finite-horizon rate.

This calculation minimizes a fixed-observation norm upper bound, not the actual MSE. The actual population discount bias is the signed finite sum

$$\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta) = \sum_{n=1}^{N} \sum_{r=1}^{n-1} (1-\alpha^r) \Delta_{n,r}, \tag{S326}$$

whose first-order coefficient can cancel. Consequently, Equation (S325) is not a lower bound on the optimally tuned MSE and does not rule out an interior α with smaller MSE than both endpoints.

Remark 3 (Fixed lag and geometric memory)**.** *For $0 \le k < N$, the fixed-lag score and its population truncation error are, respectively,*

$$\sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_s) + m(\xi_N) \left(\sum_{s=N-k}^{N} h_s \right), \tag{S327}$$

$$\sum_{s=1}^{N-k-1} \sum_{u=s+k+1}^{N} \Delta_{u,u-s}. \tag{S328}$$

Lemma S10 therefore gives

$$\left\| \mathrm{Bias}_k^{\mathrm{pop}} \right\|_2 \le C(N-k-1)\sqrt{p}\, G'(\theta) \frac{\varrho^{k+1}}{1-\varrho}. \tag{S329}$$

In the neutral model of Equation (S321), direct multinomial calculation gives

$$\begin{aligned} &\mathrm{Var} \left\{ \sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_s) + m(\xi_N) \left(\sum_{s=N-k}^{N} h_s \right) \right\} \\ &= \sigma^2 \left[(N-k-1)\{1-(1-J^{-1})^{k+2}\} + \sum_{a=0}^{k} \{1-(1-J^{-1})^{a+2}\} \right]. \end{aligned} \tag{S330}$$

This has order $N(k+1)/J$ when $1 \ll k \ll \min(N, J)$. A general particle-variance bound of that order requires the joint-lineage estimate identified after Lemma S13 and is not asserted here.

The scalar geometric-discount bias bound is an exact geometric mixture of the fixed-lag scalar bounds:

$$\sum_{r=1}^{\infty}(1-\alpha^r)\varrho^r = \sum_{q=0}^{\infty}(1-\alpha)\alpha^q \frac{\varrho^{q+1}}{1-\varrho} = \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}. \tag{S331}$$

If $k = \lfloor \alpha/(1-\alpha) \rfloor < N$, the ratio of this bound to $\varrho^{k+1}/(1-\varrho)$ satisfies

$$\frac{1}{2} < \frac{(1-\alpha)\varrho^{-k}}{1-\alpha\varrho} \le \varrho^{-k}. \tag{S332}$$

Thus the two scalar bounds are within fixed factors when $k|\log \varrho| = O(1)$. This comparison does not turn geometric discounting into a hard cutoff: from the exact signed representation,

$$\lim_{\alpha \uparrow 1} \frac{\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta)}{1-\alpha} = \sum_{n=1}^{N}\sum_{r=1}^{n-1} r\Delta_{n,r}, \tag{S333}$$

and the assumptions do not force the right side to vanish. Hence the generic geometric-discount population bias is first order in $1-\alpha$, even though its coefficient contains the forgetting powers ϱ^r.

S8 Figures for Bayesian Inference

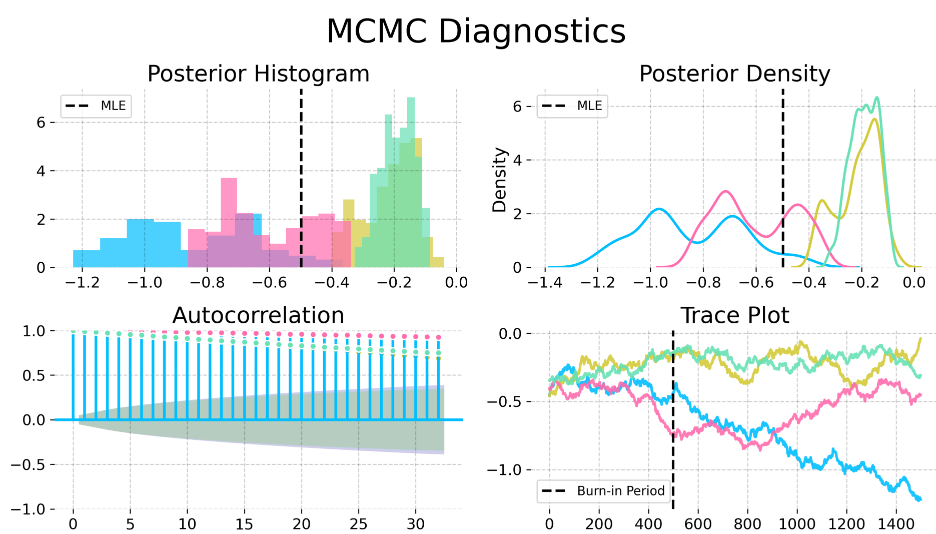

Figure S-1: Convergence diagnostics for the Metropolis-Hastings variant particle MCMC with the random walk proposal and an informative empirical prior from IF2, on the trend parameter in the Dhaka cholera model of King et al. (2008). The poor convergence shown here, using an MCMC algorithm that does not have access to derivatives, contrasts with Fig. 6 in the main text.

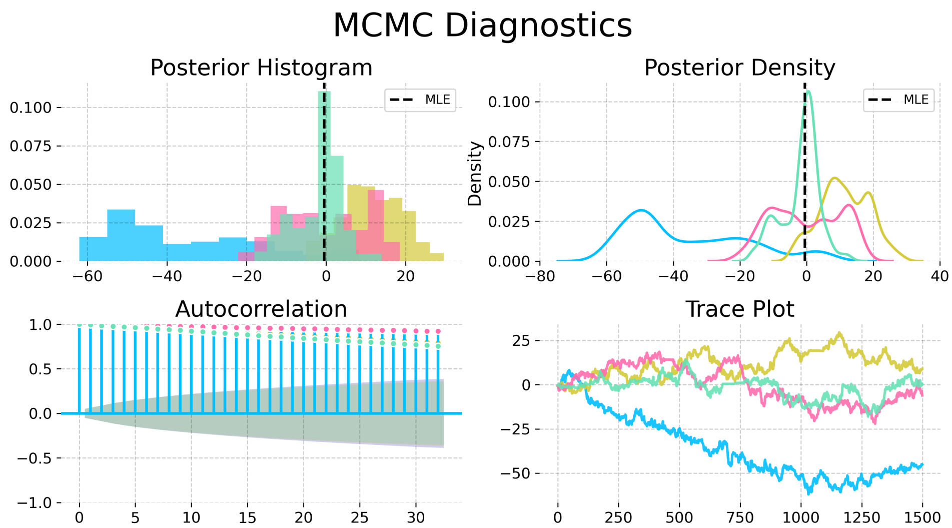

Figure S-2: Convergence diagnostics for a No-U-Turn Sampler (NUTS) with uniform priors on a compact set, on the trend parameter in the Dhaka cholera model of King et al. (2008). The NUTS sampler explores more of the posterior than the Metropolis-Hastings sampler, but fails to converge quickly. This poor convergence, compared with Fig. 6 in the main text, demonstrates the value of the empirical prior for obtaining successful convergence.

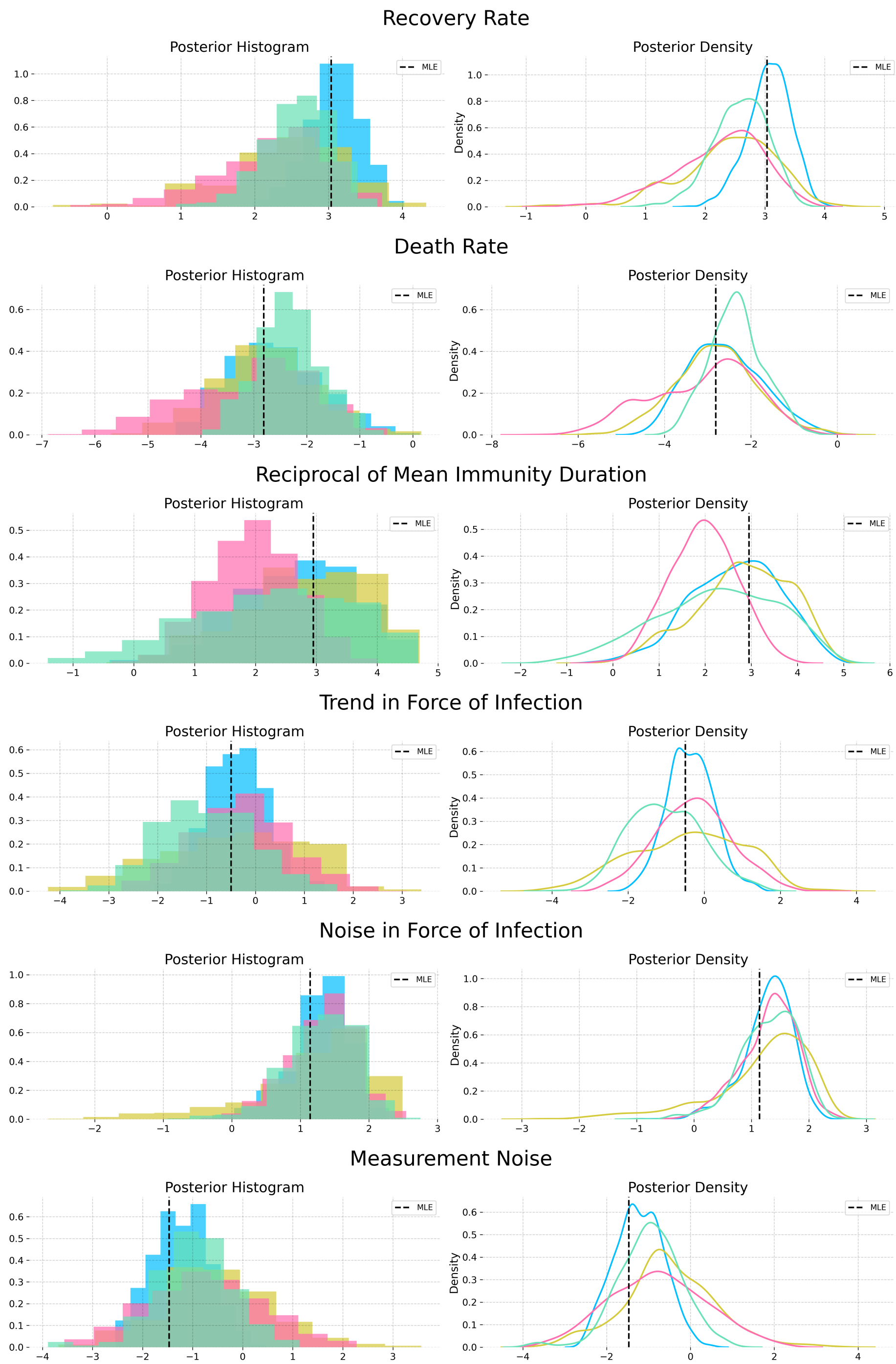

Figure S-3: Posterior estimates from NUTS powered by MOP-α with the informative nonparametric empirical prior obtained from a kernel density estimator on the IF2 parameter cloud. The posterior estimates from each chain are largely in agreement.

S9 Justification for the Choice of Baseline Comparisons

The reasoning we use in the main text to support the magnitude of our contribution is as follows: (i) iterated filtering methods are the current state of the art, and have been for a decade; (ii) we have demonstrated a substantial advance over iterated filtering. The main text focuses on demonstrating (ii), while supporting (i) via references. Listing a few references amounts only to anecdotal evidence, which is not sufficient to persuade someone properly skeptical about the claim. Therefore, this supplement provides an additional quantitative evaluation.

We count the number of papers published in three top scientific journals (*Science*, *Nature*, and *Proceedings of the National Academy of Sciences of the USA (PNAS)*) that carry out statistical inference for partially observed nonlinear non-Gaussian stochastic dynamic models to address a scientific question by fitting mechanistic models to time series data. For all methods, papers in these high-profile journals are expected to represent only be the tip of the iceberg, but this count provides a concrete measure of the capability of the methods to handle world-class scientific inference problems. We could have considered additional journals, but the three we chose are undoubtedly prestigious and their scope covers the full range of scientific endeavor so there is less room for subfield bias. For iterated filtering, our high-impact application count gives the following: 1. He et al. (2023), 2. Fox et al. (2022), 3. Subramanian et al. (2021), 4. Pei et al. (2021), 5. Li et al. (2020), 6. Giezendanner et al. (2020), 7. Wale et al. (2019), 8. Pons-Salort and Grassly (2018), 9. Ranjeva et al. (2017), 10. Martinez et al. (2016) , 11. Becker et al. (2016), 12. Bakker et al. (2016), 13. Laneri et al. (2015), 14. Blake et al. (2014), 15. Blackwood et al. (2013a), 16. Blackwood et al. (2013b), 17. King et al. (2008),

A referee advised consideration of an alternative approach by Gu and Kong (1998) that we found to have a high-impact application count of zero, via Google Scholar.

A referee also recommended comparison with Lindsten et al. (2014), for which the high-impact application count is also zero. Lindsten et al. (2014) has been employed once in *Nature Communications* (Calafat et al., 2018) though that journal did not quite make it onto our elite list. We also found five applications of iterated filtering in *Nature Communications* (Ranjeva et al., 2019; Rodó et al., 2021; Santos-Vega et al., 2022; Briga et al., 2024; Perez-Saez et al., 2025). There is a paper in the Statistics section of *PNAS* by Wang et al. (2022) that demonstrates the method of Lindsten et al. (2014), but here we are counting the demonstrated ability of the methodology to generate high-profile scientific papers, not statistical methodology papers. Otherwise, we would also count Ionides et al. (2006) and Ionides et al. (2015) for iterated filtering.

This reasoning supports iterated filtering as an empirically justified point of reference, in preference to the approaches of Lindsten et al. (2014) and Gu and Kong (1998). Further, the method of Lindsten et al. (2014) does not have the plug-and-play property so it cannot readily be applied to our example. The method of Gu and Kong (1998) would have to be combined with an approach such as Andrieu et al. (2010) that has known scalability problems. Therefore, we focus on comparing with iterated filtering and we do not provide a comparison with Lindsten et al. (2014) or Gu and Kong (1998).

Benchmark tasks are invaluable for making progress on challenging problems (Donoho, 2017). The Dhaka cholera data and model have served as such a benchmark since its introduction by King et al. (2008). Other papers testing methods on this example include Ionides et al. (2015); Fasiolo et al. (2016); Wycoff et al. (2026). Thus, our test is established as a demanding inference problem that challenges the best available methods. We note that this benchmark challenge contains various difficulties that arise commonly in scientific inquiry but are absent from simple SIR models such as the simulation study considered by Lindsten et al. (2014). Ability to accommodate such features is critical for the methodology to be effective for scientific inquiry.

Iterated filtering has recently been used as a benchmark for new POMP model methodology by Whitehouse et al. (2023). This analysis presented numerical results that claimed to beat iterated filtering, but was subsequently found to contain numerical errors. Corrected results are favorable for iterated filtering methods (Hao et al., 2024; Whitehouse et al., 2025; He et al., 2025).

Recent results of Chen et al. (2025) have extended the previous theoretical understanding of iterated filtering. The ongoing study of iterated filtering as a state-of-the-art approach to inference for POMP models gives additional support for our decision to use iterated filtering as a benchmark for our new methodology.

S10 Algorithmic Parameters

We include the full set of algorithmic parameters used for the Dhaka results in Table S-1. The warm start iterations are IF2 iterations, which are followed by gradient descent (GD) iterations when using IFAD. The IF2 cooling rate is the fractional reduction in the parameter perturbations over 50 iterations, so that a cooling rate of 0.5 corresponds to approximately 1% cooling per iteration. The random walk standard deviation (RWSD) scales the noise added at each filtering step to each regular parameter. Initial value parameters (IVPs), that affect the value of the latent process only at time t_0, are treated differently: they are perturbed only at time t_0, with a larger RWSD. Parameters are transformed to a unit scale, for example by taking logarithms or a logistic transformation, so that a single RWSD value works for all parameters.

Hyperparameter	**Standard IF2**	**IFAD-0**	**IFAD-0.97**	**IFAD-1**
Number of Particles (J)	5,000	5,000	5,000	5,000
Warm-start Iterations	650	60	175	60
GD Iterations	—	225	175	225
IF2 Cooling Rate	0.8	0.5	0.5	0.5
Initial RWSD	0.02	0.02	0.02	0.02
Initial RWSD (IVPs)	0.16	0.16	0.16	0.16
Gradient Optimizer	—	Adam	Adam	Adam
Momentum (β_1)	—	0.0	0.9	0.9
LR Schedule	—	Cosine Decay	Cosine Decay	Cosine Decay
Initial LR (η_0)	—	0.1	0.1	0.1
Warm-up Steps	—	10	10	10
Eval. Particles / Reps	5,000 / 36	5,000 / 36	5,000 / 36	5,000 / 36

Table S-1: Algorithmic parameters for the global search benchmarking on the Dhaka cholera model.

Following some experimentation, IFAD was implemnenteed by plugging the DMOP gradient estimate into the Adam optimizer (Kingma and Ba, 2014). The momentum parameter in Adam was turned off for IFAD-0, since this low-variance estimate did not reqquire it. Decreasing the learning rate (LR) was found to be helpful, and we selected the commonly used cosine decay. Warm-up steps were taken with Adam to stabilize its initial performance and avoid noisy

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